

A multilayer integrative approach for diagnosis, classification and severity detection of electrical faults in photovoltaic arrays

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ABSTRACT

Early detection of faults in photovoltaic (PV) systems is crucial for improving their lifespan and reliability. Conventional protection devices may not be able to detect electrical faults under critical conditions. This usually occurs due to (i) the current-limiting nature and non-linear output characteristics of PV arrays, (ii) the changing environmental conditions (such as irradiance and temperature variations), and (iii) the influence of the maximum power point tracker (MPPT), particularly in the presence of critical fault impedance values or mismatch levels. Therefore, modern data-driven methods are required for fault detection and classification in PV arrays. However, careful investigation in previous studies reveals the existing gaps and limitations, such as poor accuracy, and incomprehensive models which have not considered various electrical faults, low mismatch levels, and critical fault impedance values. To this end, in this study, a comprehensive multilayer model is proposed which consists of six layers for diagnosis, classification and severity identification of electrical faults in PV arrays. Each layer adopts a weighted ensemble learning (WEL) algorithm consisting of three classifiers namely Support Vector Machine (SVM), Naïve Bayes (NB) and Logistic Regression (LR). Besides, a genetic algorithm (GA) is employed in each layer to find the optimal weights for each classifier (i.e., SVM, NB and LR). The initial dataset is acquired through an investigation into the PV array current–voltage (I–V) characteristic curve and extraction of several features under various faulty and normal conditions. To reduce the dataset dimensionality thus computationally simplifying the training process, the sequential floating forward selection (SFFS) algorithm is utilized in each layer as a powerful feature selection technique. The results show a highly accurate performance in fault diagnosis, classification, and severity assessment, with an average simulation and experimental accuracies of 98.9% and 98.37%, respectively.

1. Introduction

Solar energy has attracted an immense attention in recent decades due to its advantages, such as worldwide accessibility, and ease of installation. The cumulative photovoltaic (PV) installations exceeded 1 TW at the end of 2022 and went beyond 1.5 TW by the end of 2023 (Aghaei et al., 2022). However, PV components are constantly affected by environmental and climatic factors, such as shades, hail, lightning, snow, dirt, dust, harsh sunlight, heat and cold, moisture, corrosive gases, etc. as they operate in a natural environment (Santhakumari & Sagar, 2019). This results in various failures which affect the normal operation, the service life and the efficiency and in some cases may even cause fire

hazards (Onal & Turhal, 2021). Therefore, comprehensive and careful monitoring of PV components is becoming increasingly critical to early detect and clear the faults (Berghout et al., 2021; Emamian et al., 2022; Vieira et al., 2022). Conventional protective devices such as overcurrent protection devices (OCPDs) and ground fault detection interruptors (GFDIs) are unable to perform properly under specific conditions due to non-linear output characteristics of PV arrays, PV current-limiting nature, critical fault impedances, low irradiance conditions, and the presence of MPPTs (Flicker & Johnson, 2013). Therefore, modern and automatic methods are inevitably required.

In recent years, several modern methods and techniques based on artificial intelligence (AI), specifically machine learning (ML) which is

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the main subset of AI have been utilized to detect and classify various faults in PV systems (Suliman et al., 2024). A novel state-of-the-art fault detection scheme based on internet of things (IoT) technology is proposed in (Hojabri et al., 2022). In another study (Abbas & Zhang, 2021), authors proposed a hybrid fault detection model based on a combination of neural networks and fuzzy inference system. Also, in (Djalab et al., 2020) a sophisticated three-layered ANN-based model is presented for precise fault detection in solar panels. However, all the above mentioned models do not seem reliable enough since they are only capable of detecting severe faults in PV arrays. Note that severe faults are easy-to-detect events even using conventional protection devices. Therefore, they disregarded the most critical conditions in PV arrays fault detection process i.e., when a critical number of modules (usually a small number) is involved in a fault occurrence and/or the fault contains a critical impedance value. Critical conditions can create an anomalous status extremely similar to PV array normal operating condition thus making the fault detection process incredibly difficult (or sometimes impossible) (Daher et al., 2023; Aghaei et al., 2022).

On the other side, several studies in literature have attempted to fill the previous gap by considering a number of critical conditions in the process of fault detection. However, several shortcomings can still be noticed in their models. In various studies such as (Amiri, Kichou, et al., 2024; Hajji et al., 2023; Hichri et al., 2024; Mustafa et al., 2023; Yahyaoui et al., 2023), in order to train the proposed models, a massive amount of dataset is required. However, data acquisition tends to be extremely demanding especially under harsh weather conditions. Although the presented models in (Dhimish & Tyrrell, 2023) and (Badr et al., 2021) have considered both a small dataset and a number of critical conditions in fault detection, the downside is the models insufficient accuracies which noticeably reduces the models reliability. Moreover, authors in (Amiri, Oudira, et al., 2024) have presented a seemingly accurate model which considers a number of critical conditions in fault detection and is trained using a relatively small dataset. However, the model is incomprehensive since the capability of fault detection is only limited to a few kinds of faults in PV arrays. Accordingly, an increase in the number of faults in the dataset will adversely affect the model accuracy and reliability in real-world conditions (Aghaei, 2022).

To fill the above mentioned gaps in literature, in this study, challenges are addressed and a comprehensive method is proposed for simultaneous diagnosis and classification of all types of electrical faults at the DC side of PV arrays. To this end, first, a comprehensive examination of the PV array current-voltage (I-V) characteristic curve is conducted under various conditions including line-line (LL), line-ground (LG), open-circuit (OC), string degradation, and array degradation faults, as well as normal operating condition to construct the initial dataset. Then, several features are extracted from the initial dataset using a number of specific predefined points on the PV array I-V curves. A comprehensive multilayer model is then proposed to detect and classify various faults and assess the severity of LL and LG faults especially under critical conditions. The model consists of six layers, each of which is assigned a carefully defined responsibility. Furthermore, to obtain the best performance, a weighted ensemble learning (WEL) model consisting of three ML classifiers which are Support Vector Machine (SVM), Naïve Bayes (NB) and Logistic Regression (LG) is applied in each layer. A genetic algorithm (GA) is utilized in WEL model to optimize the weights assigned to each classifier. Moreover, the sequential floating forward selection (SFFS) algorithm is used in each layer as a feature selection technique to select the most relevant features, which helps to improve the performance of the model by reducing the amount of required training data and providing a computationally simple training process.

Therefore, in light of the above considerations, the present paper makes the following novelties and contributions

- Unlike a large number of previous studies which required a massive amount of data to provide fully-trained flawless models which causes various problems, such as model complexity, data acquisition challenges, data storing problems, etc., a new approach is used in this paper to detect, classify, and assess the severity of various faults in PV arrays which is proven to be so capable and powerful to work only using a small dataset either in the simulation process or the real-world experiments. This definitely minimizes the risk of data acquisition in real-world condition which is demanding, energy-consuming, time-consuming, and in some cases impossible.
- Numerous studies in literature have attempted to detect only severe faults in PV arrays which are the easiest conditions in the process of detection even using conventional protection devices. Therefore, they disregarded the critical fault detection conditions which are the most challenging conditions in the process of detection. However, in this paper, critical conditions in fault detection are fully investigated through the severity assessment various faults under critical mismatch levels and/or critical impedance values which make the process of fault detection extremely difficult.
- Literature of PV array fault detection is replete with incomprehensive models which are capable to detect only a few kinds of faults, thus casting doubt on their practicability, especially when implemented in real-world conditions. In this regard, the present paper proposes a comprehensive multilayer model to cover all frequent electrical faults in PV arrays thus making it more pragmatic to be implemented on real PV systems.
- The accurate performance of the model is guaranteed through combining ML techniques and extracting the final result using a weighted ensemble learning (WEL) algorithm with a genetic algorithm (GA) to provide a precise weight optimization. The proposed multilayer model has been able to detect, classify, and assess the severity of various faults with high accuracy of 98.9 % in the simulation process and 98.37 % in a real-world experiment. The proposed model has an advantage over the existing ML techniques in the literature due to its high accuracy which is produced using only a small number of data samples.
- Complex models including datasets containing high dimensions usually fail to be implemented in real-world conditions. To address the problem, in this paper, the SFFS algorithm is presented as a feature selection technique to select only the most effective features in each layer thus reducing the dataset dimensionality and providing a computationally simple training process.

The rest of the article is arranged as follows: Section 2 introduces all frequent electrical faults in PV arrays and discusses the protection challenges, their characteristics and their possible impacts PV array power generation. The proposed multilayer method is presented in Section 3. In Section 4, simulation results are provided which is followed by the experimental results and discussion in Section 5. Finally, Section 6 wraps up the paper by a brief summary of the whole process and potential future work.

2. Faults analysis in PV arrays

2.1. PV array configuration

Fig. 1 depicts a typical configuration of a PV array which is comprised of connection cables, current and voltage sensors and protective devices, such as OCPDs, and GFDIs (in grounded systems). To set up a PV array, PV modules are arranged in series and parallel to yield the desired level of current and voltage, thus the desired output power.

In this study, a stand-alone 10 (series) $\times$ 3 (parallel) typical PV array is designed in MATLAB/Simulink with the nominal output power of 4.5 kW under standard test conditions (STC: irradiance = 1000 W/m², temperature = 25 °C, airmass = 1.5) to analyze the impacts of the electrical faults on the PV array I-V curve. The PV array is implemented

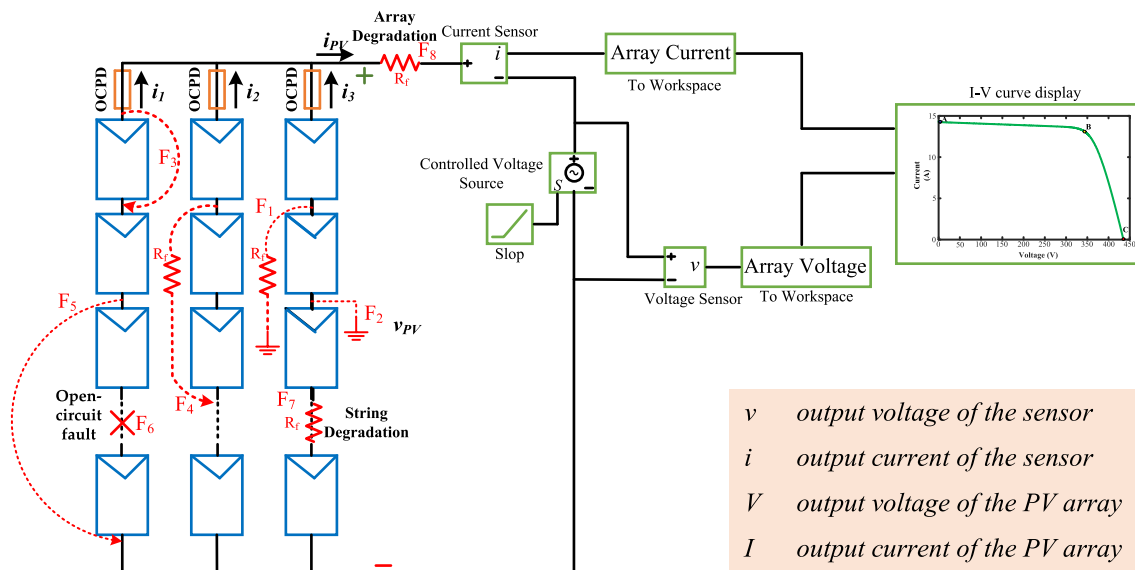


Fig. 1. A typical 10×3 stand-alone PV configuration.

in MATLAB version R2016b and simulations are run using a Laptop with Microsoft Windows 10 which uses an Intel Corei7-4710 MQ @2.50 GHz processor with 8 GB of RAM. [Table 1](#) lists the main parameters of the PV module which is used in the process of simulation.

2.2. Faults analysis in PV arrays

PV arrays are prone to various faults, such as LL, LG, OC, and degradation faults. In case the faults remain undetected in PV arrays, they can affect the output power, the performance of the system, the reliability and lifetime of the system, and even have the potential for catastrophic events, such as fire hazards. In this section, a detailed analysis of various frequent electrical faults in PV arrays, the characteristics, protection challenges and impacts on I-V curve is carried out for a better understanding of the necessity of fault detection and severity identification ([Khalil et al., 2020](#)).

2.2.1. Line-to-Ground (LG) faults

A line-to-ground (LG) fault occurs when there is an unexpected connection between earth-ground conductors (EGCs) and current-carrying conductors (CCCs) in a PV array, leading to a continuous flow of current into the ground. The EGCs are used to protect against electric shocks (Mellit et al., 2018). LG faults can be categorized as 1) an accidental short circuit between a normal conductor and an EGC in a module junction box, 2) cables insulation breakdown which usually occurs due to the chewing of cable insulation by the rodents and 3) a short circuit between a PV cell and the grounded module frames (Zhao & Lyons Jr, 2011). In Fig. 1, F1 and F2 show two LG fault cases where the former has occurred with a high impedance (R_f) and a mismatch level of 90 % while the latter is directly connected to the ground with 80 % of mismatch.

LG faults are able to create dramatic impacts on PV array I-V curves.

Table 1
The main parameters of the PV module in simulation process.

Parameter	Value
Power at maximum power point (P_{MPP})	150 W
Open circuit voltage (V_{OC})	43.5 V
Voltage at maximum power point (V_{MPP})	34.5 V
Short circuit current (I_{SC})	4.75 A
Current at maximum power point (I_{MPP})	4.35 A
Number of series cells in each module (N_s)	72

Fig. 2 illustrates the changes which occur due to an LG fault on an I-V curve using six operation points (i.e., points A-F). During the occurrence of an LG fault, the pre-fault array and string operation points which are A and D move instantaneously to points B and E, respectively, where the array current is zero and the string current is negative. In the meanwhile, the MPPT begins to relocate the operating points to obtain the maximum output power by reducing the voltage in the PV array which forces the post-fault array operation point to C. The operating point of the faulted string is shifted to F, leading to a positive return of the current. Therefore, a significant decrease in output power can be witnessed due to the change in the operating point. As a result, it can be concluded that a simultaneous decline in both output power and voltage at the operating point indicates the presence of an LG fault in PV systems.

Protection against LG faults may sometimes seem challenging in that 1) the MPPT can drop the fault current magnitude and thus make it less possible for the OCPDs and GFDIs to function properly, and 2) GFDIs are not sensitive enough to detect a high-impedance LG fault.

2.2.2. Line-Line (LL) faults

The term “line-line (LL) fault” refers to a short circuit that occurs between two points in a PV array with different voltage levels, which are commonly referred to as mismatch levels. LL faults may occur due to three possible causes: 1) there might be an accidental short circuit between two CCCs, 2) cables insulation might face serious breakdown and 3) there might happen an internal LL fault in DC junction boxes due to several causes, such as mechanical damages, water ingress, and corrosion (Zhao et al., 2013). In Fig. 1, F3 depicts an LL fault with 10 % of mismatch as only one PV module out of ten is involved. Accordingly, F₄ and F₅ deal with 20 % and 80 % mismatch percentages, respectively. Note that F₄ demonstrates an LL fault occurrence under a high impedance value (R_f) as well.

Fig. 3(a) illustrates I-V curves under STC and different states of LL faults, one with 10 % of mismatch and a 10-ohm impedance, one with 20 % of mismatch without an impedance and one with 30 % of mismatch and again a 10-ohm impedance. LL faults can also be difficult and seriously misleading to detect. As shown in Fig. 3(a), this is due firstly to the fact that LL faults under low mismatch percentages and high impedance values seem to be extremely similar to PV array normal condition. Secondly, LL faults with low mismatch levels but low impedance values tend to appear fairly close to those with high mismatch percentages and high impedance values.

Furthermore, similar to LG faults, undetected LL faults can result in

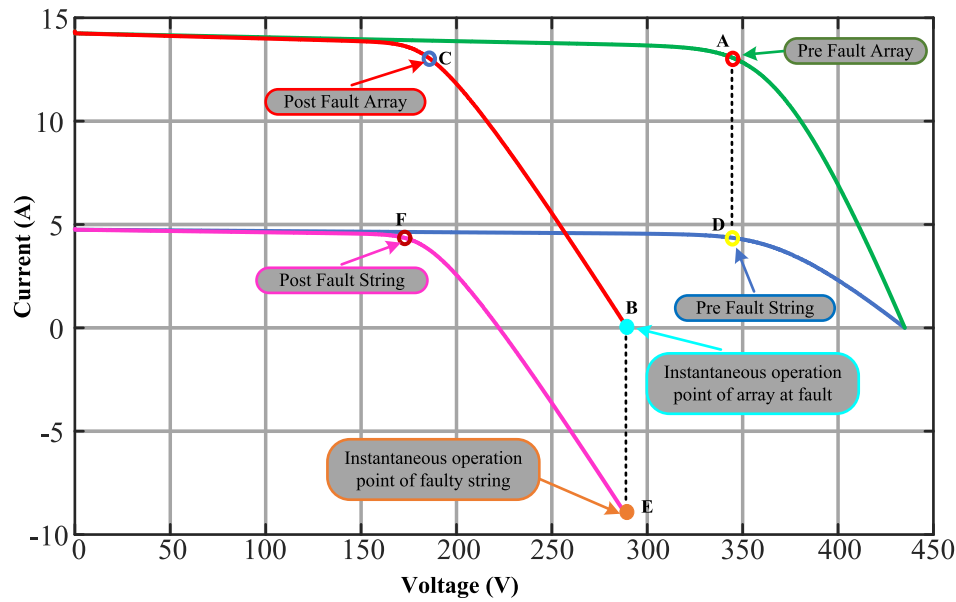


Fig. 2. Effects of an undetected LG fault on I-V curve.

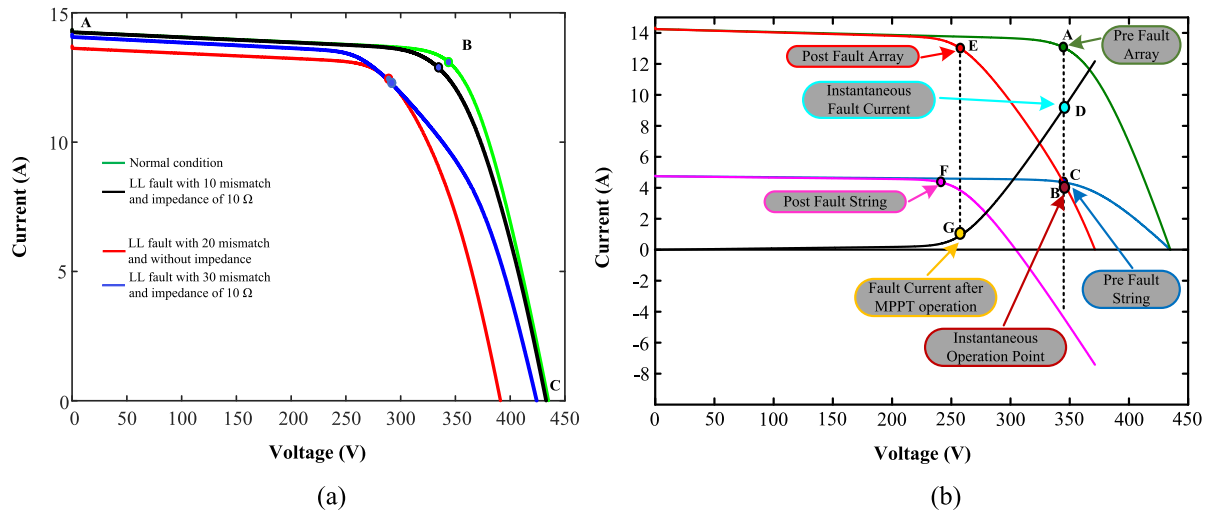


Fig. 3. (a) Severity of an LL fault based on different mismatch percentages, with and without an impedance (b) Effect of an undetected LL fault on I-V curve.

significant changes on I-V curves, as shown in Fig. 3(b). While the MPPT can mitigate the damaging effects of LL faults by shifting the operating point of the PV array from point A to point E, this often results in a substantial decrease in power, voltage, and fault current. On the other hand, protection devices may fail to accurately detect LL faults especially in three scenarios: 1) when the voltage difference between the two ends of the line is less than 20 %, the fault current may be significantly reduced and mislead the OCPDs, 2) as previously discussed, the presence of the MPPT can significantly lessen the fault current magnitude, hence, LL faults may remain undetected for an indefinite period of time, and 3) under low irradiance levels and high impedance conditions, the LL fault current is still extremely low even for a 40 % mismatch level (Sabbaghpur Arani & Hejazi, 2016).

2.2.3. Open-Circuit (OC) faults

An open circuit (OC) fault in PV arrays refers to an accidental break in the current-carrying conductors, which can occur due to cracks in PV cells, PV modules or module connections, usually wiring and junction boxes. In Fig. 1, F6 demonstrates a typical OC fault which may have happened due to any of the previously mentioned reasons. It is assumed

that F_6 has happened under STC, so it does not generate a large amount of fault current. As one out of three strings is now disconnected, over a third of the output power is unavoidably lost. Also, the voltage at the operating point remains largely constant both during and after the fault. Thus, the faulted PV array quickly reaches its optimum operating point. Generally, OC faults may occur as a result of 1) cyclic thermal stress, 2) hail or wind loading and 3) damage during processing and assembly (Sabbaghpur Arani & Hejazi, 2016).

In OC faults, both current and voltage at MPP are affected. Besides, further disconnected strings lead to less PV array short circuit current such that in case n parallel strings in a PV array are involved in an OC fault, $n \times I_{sc}$ will be subtracted from the whole PV array short circuit current where I_{sc} denotes the short circuit current in each string (see Fig. 4).

2.2.4. Degradation faults

Despite being reliable energy sources, PV modules and arrays are highly prone to degradation faults particularly after a prolonged period of time. PV modules can be degraded due to 1) long-term exposure to ultraviolet radiation (known as optical degradation), 2) an increase in

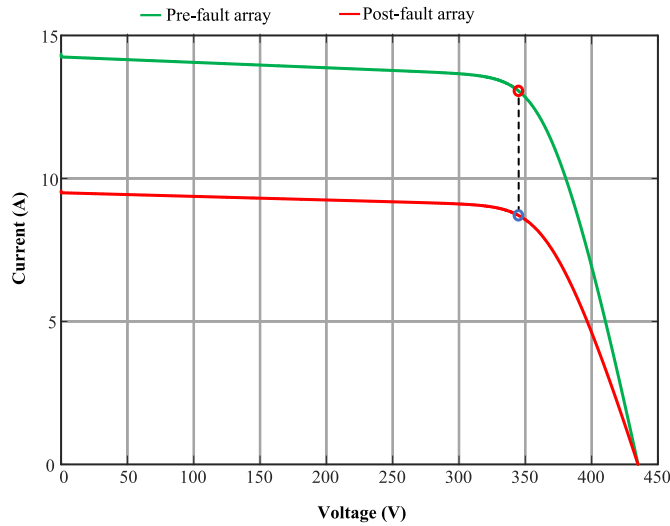


Fig. 4. Effect of an OC faults on I-V curve.

series resistance or a decrease in either shunt resistance or short circuit current (known as cell degradation) and 3) mismatched cells as a result of surface soiling, cell cracking, partial shading and so forth (Sabbagh-pour Arani & Hejazi, 2016).

Since degradation faults have already been investigated in many previous papers such as (Katayama et al., 2019; Pei et al., 2020), this study is only limited to the influence of the series resistance. In addition, both string and array degradation faults are explored by adopting series resistances embedded both in a string and in the output of an array, as shown in Fig. 1, F7 and F8, respectively.

The effect of degradation faults on the PV string I-V curve with several impedance values is demonstrated in Fig. 5(a). As depicted in Fig. 5(a), even with the presence of a relatively high 12-ohm resistance, the I-V curve is very similar to that of a normal condition. On the other hand, in Fig. 5(b) which shows the impacts of PV array degradation faults on the I-V curve using different series resistance values, it is evident that as the series resistance increases, the curve displays a very different behavior from that of a normal condition.

2.3. Feature extraction process

To improve the training process and providing the ML model with a better interpretation of the dataset, feature extraction seems a crucial procedure. In this study, sixteen features are extracted from the initial dataset obtained from five specific points on the I-V curve (see Fig. 6) and from the module datasheet. Table 2 describes the five points in detail. The extracted features are also listed in Table 3. As can be seen in

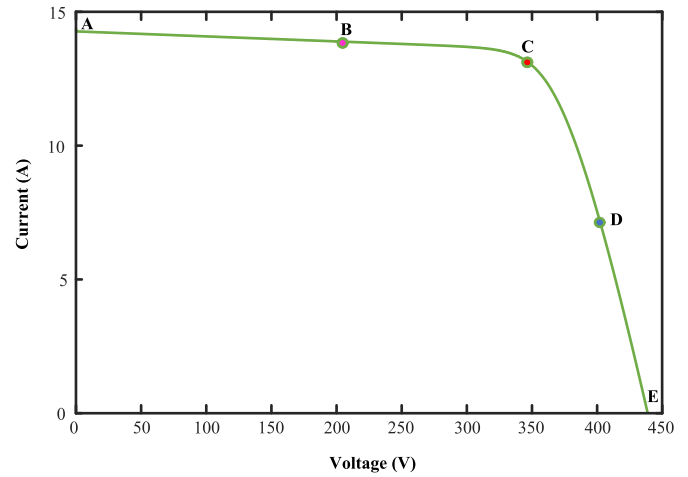


Fig. 6. The five specific points on the I-V curve for extracting the features.

Table 2

Initial data used in the proposed method.

Source	Data type	Description
I-V curve	A: (0, I_{SC})	The point refers to the short-circuit current on I-V curve.
	B: ($V_{OC}/2$, $I_{VOC}/2$)	The point refers to the half of open-circuit voltage on I-V curve.
	C: (V_{MPP} , I_{MPP})	The point refers to the maximum power on I-V curve.
	D: ($V_{ISC}/2$, $I_{SC}/2$)	The point refers to the half of short-circuit current on I-V curve.
	E: (V_{OC} , 0)	The point refers to the open-circuit voltage on I-V curve.
PV module datasheet	I_{SC} (STC)	The short-circuit current under STC
	V_{OC} (STC)	The open-circuit voltage under STC
	I_{MPP} (STC)	The current at maximum power point under STC
	V_{MPP} (STC)	The voltage at maximum power point under STC

Table 3, the first six features deal with the current and voltage changes in five above-mentioned points in the I-V curve. The seventh feature indicates the rate of conductivity change in a PV array. The next two features refer to the ratio of V_{MPP} and I_{MPP} changes to V_{OC} and I_{SC} , respectively. The tenth to fifteenth ones show slope changes on the I-V curve in five specified points and eventually the last feature defines the normalized fill factor.

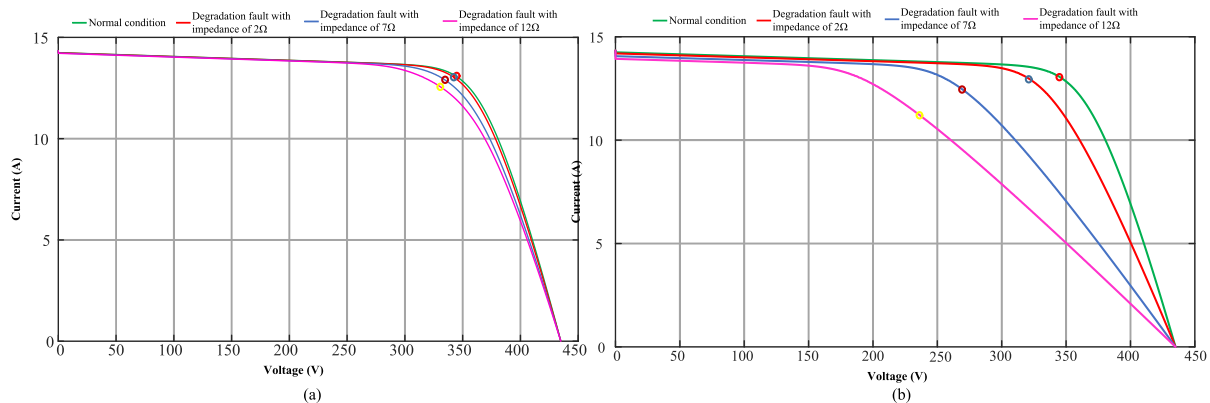


Fig. 5. Effect of a degradation fault on I-V curve with different impedances (a) a string degradation (b) an array degradation.

Table 3

Features extracted from PV array I-V curves.

$f_1 = I_{SC}/I_{SC(STC)}$	(1)	$f_9 = f_4/f_1$	(9)
$f_2 = V_{OC}/V_{OC(STC)}$	(2)	$f_{10} = (I_{MPP} - I_{SC})/(V_{MPP})$	(10)
$f_3 = V_{MPP}/V_{MPP(STC)}$	(3)	$f_{11} = (-I_{MPP})/(V_{OC} - V_{MPP})$	(11)
$f_4 = I_{MPP}/I_{MPP(STC)}$	(4)	$f_{12} = (I_{V_{OC}} - I_{SC})/(\frac{V_{OC}}{2})$	(12)
$f_5 = \frac{I_{V_{OC}}}{2}/I_{SC(STC)}$	(5)	$f_{13} = (I_{MPP} - I_{V_{OC}})/(\frac{V_{MPP} - V_{OC}}{2})$	(13)
$f_6 = \frac{V_{I_{SC}}}{2}/V_{OC(STC)}$	(6)	$f_{14} = (\frac{I_{SC}}{2} - I_{MPP})/(\frac{V_{I_{SC}} - V_{MPP}}{2})$	(14)
$f_7 = f_4/f_3$	(7)	$f_{15} = (\frac{I_{SC}}{2})/(V_{OC} - \frac{V_{I_{SC}}}{2})$	(15)
$f_8 = f_3/f_2$	(8)	$f_{16} = \frac{FF}{FF_{(STC)}} \rightarrow FF = \frac{V_{MPP} \times I_{MPP}}{V_{OC} \times I_{SC}}, FF_{(STC)} = \frac{V_{MPP(STC)} \times I_{MPP(STC)}}{V_{OC(STC)} \times I_{SC(STC)}}$	(16)

3. The proposed method

In this paper, a multilayer model is presented for detection, classification, and identification of the severity of electrical faults in PV systems. The faults considered in this study include LL, LG, OC, and degradation faults. The proposed method is a novel and intelligent ML-based model, which yields a high accuracy while it requires a small training dataset. The proposed multilayer model is comprised of six layers (see Fig. 7). The first layer is named the fault-current-based splitter (FCBS) and functions as a Yes/No binary classifier. If it outputs a “No”, the data sample proceeds to the second layer which is responsible to detect whether the sample is a normal condition or a faulty event with low fault current magnitude which behaves very similarly to a normal condition. In case the sample is distinguished as a fault event, it goes on to the third layer to be diagnosed and classified as either an OC or a string degradation fault. However, if the first layer outputs a “Yes”, the sample moves to the fourth layer to be categorized into three classes which are LL, LG and array degradation faults. Finally,

in case of an LL or an LG fault, the sample is directed towards the fifth and the sixth layer, respectively in order for its severity to be detected according to different mismatch percentages.

Additionally, previous studies have often employed a single supervised learning algorithm in the process of fault detection and classification in PV systems, despite the availability of a wide range of learning algorithms. Selecting an appropriate algorithm is a crucial and sometimes challenging step in this process, as each individual algorithm can only learn a limited number of hypotheses from the training dataset, resulting in a restricted and unstable model for fault detection and classification. To address the difficulties, a combination of three algorithms, namely Support Vector Machine (SVM), Naive Bayes (NB), and Logistic Regression (LR), is proposed in this study as an ensemble learning (EL) model. The final classification results are obtained by assigning appropriate weights to each individual algorithm. Assigning the most accurate weights to algorithms is crucial since the algorithms with better performance must possess a heavier weight than the others. This leads to an optimization problem which, in this study, is solved using a GA. Fig. 8 shows the flowchart of the proposed method. The model consists of 1) feature extraction and building up a dataset based on the I-V curve (see again Table 3), 2) data pre-processing and feature selection using the SFFS algorithm and 3) training and cross-validation. Moreover, in Fig. 8, the proposed WEL algorithm is demonstrated which is accompanied by a GA to optimize the weights. It includes 1) combining the results according to the weights optimized by a GA and 2) prediction and final model verification.

3.1. Data normalization

Data normalization is a method of standardizing data attributes while preserving their inherent properties. It is an important part of data pre-processing, especially to prevent high-value features from dominating in cases where the range of values is significantly different. Furthermore, data normalization can also improve the convergence speed of the algorithm. Among three main techniques for data normalization namely max–min, z-score, decimal scaling (Pecht & Kang, 2018), the max–min transformation is selected in this study for its linearity that keeps the relation in the original dataset. According to Eq. (17), the method squeezes the data to a closed interval between 0 and 1:

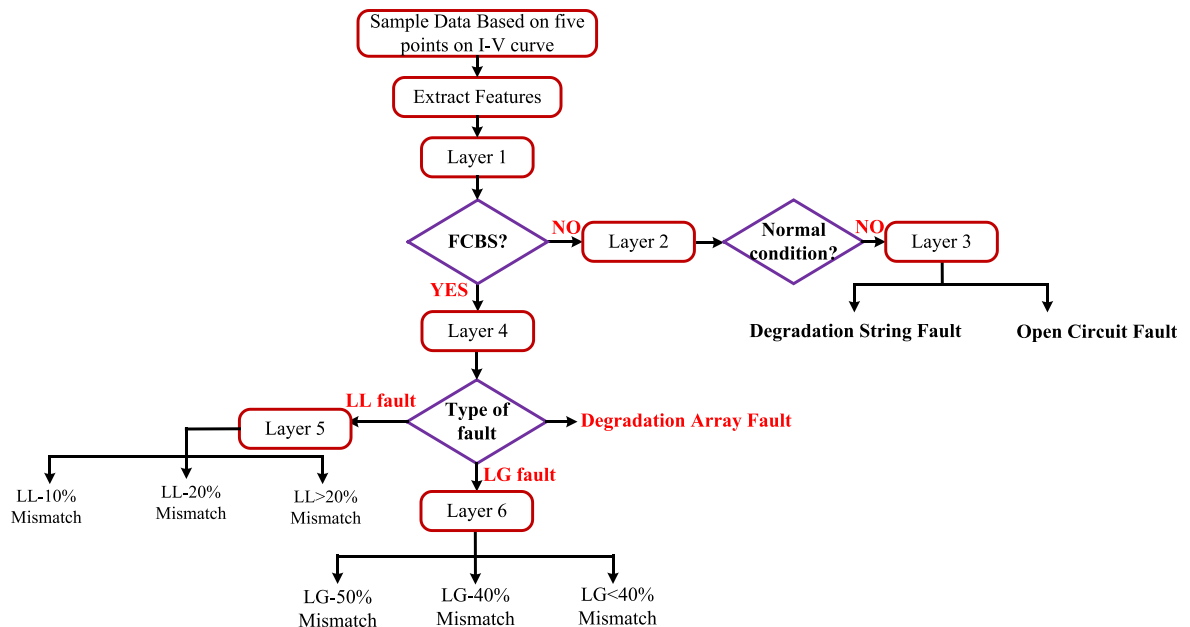


Fig. 7. The flowchart of proposed Multilayer method for electrical fault detection, classification and severity identification.

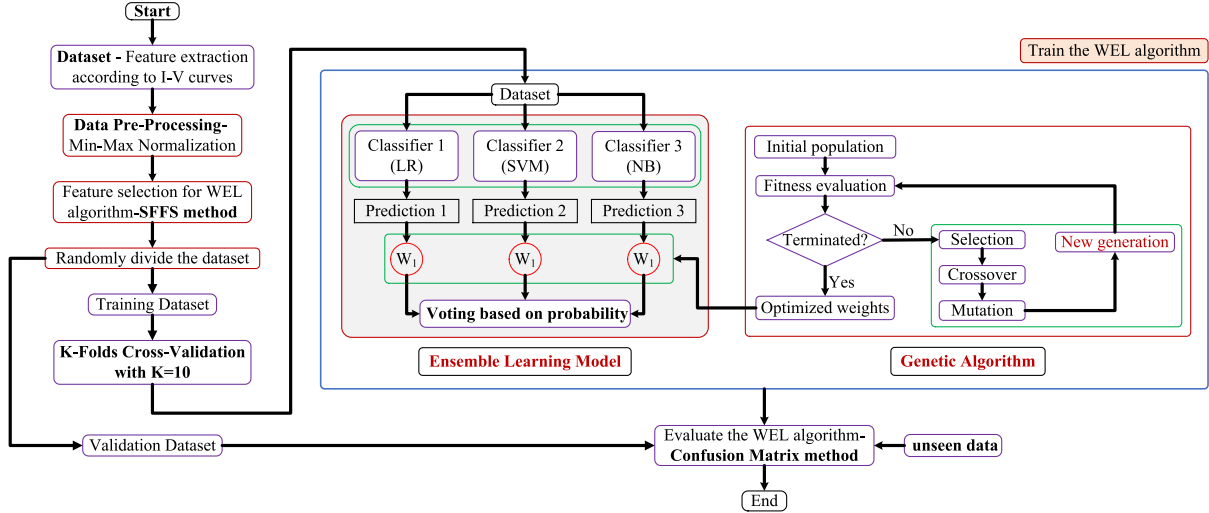


Fig. 8. The flowchart of the proposed method including the WEL algorithm and a GA.

$$a' = \frac{a - \min_B}{\max_B - \min_B} \quad (17)$$

where a is the original value of a feature B , a' is its normalized value, and $\max_B$ and $\min_B$ are the maximum and minimum values of feature B , respectively.

3.2. Feature selection process

Feature selection, also known as attribute selection, is an important step in ML processes where a subset of the extracted features is selected to improve the performance of the algorithm and avoid overfitting. By eliminating irrelevant features, the model is simplified and the resources spent on calculation and data acquisition are reduced, while the focus is on the most important features involved in the modeling process (Cai et al., 2018). Feature selection techniques can be grouped into three categories: filter, wrapper, and embedded methods. Filter methods use statistical tests to determine the correlation between the features and the target variable, independent of any machine learning algorithm. Wrapper methods, on the other hand, evaluate the usefulness of features based on the performance of the classifier. Examples of wrapper methods include sequential forward selection (SFS) and sequential backward selection (SBS). Embedded methods, like wrapper methods, aim to optimize the performance of a learning algorithm but use an intrinsic model building metric during learning. In this study, a wrapper method is used which is a combinational form of both the SFS and SBS and have avoided using them individually since 1) in the SFS, a feature, when selected, cannot be eliminated and 2) in the SBS, an eliminated feature cannot be re-selected.

For the aforesaid reasons, the sequential floating forward selection (SFFS) algorithm is deployed owing to its flexibility in that it performs backward steps after each forward step as long as the objective function increases. The flow chart of the SFFS algorithm is depicted in Fig. 9 in which k is the number of selected features in the current subset and d is the required dimensionality.

The first step is similar to SFS, beginning with an empty subset and adding newly-selected features according to the objective function. Then, if the number of selected features does not meet the required dimensionality, meaning $k \neq d$, the SFFS algorithm functions as the SBS, steps backward and conditionally leaves a feature out. If the elimination helps the objective function increase, it will be confirmed and the algorithm restarts with the new $(k-1)$ -feature subset. Otherwise, the elimination is rejected, the feature is re-added and the SFFS algorithm starts all over. The process iterates until $k = d$ or the desired function-

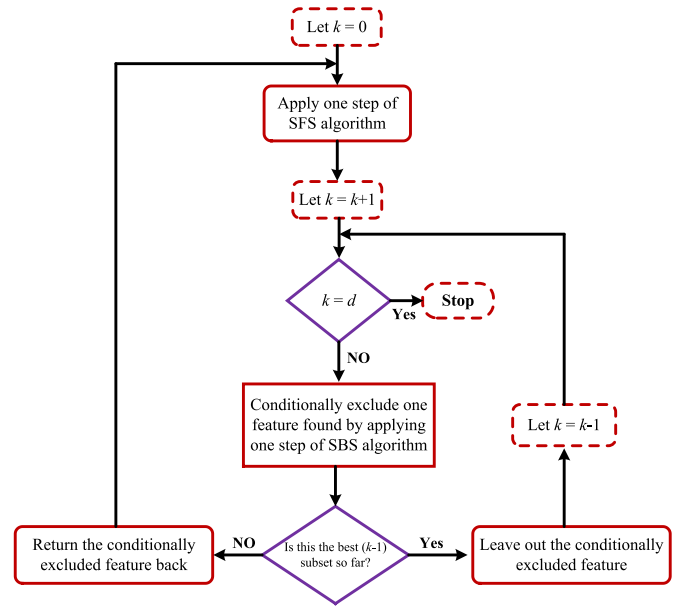


Fig. 9. The performance chart of the SFFS algorithm.

ality is obtained.

3.3. Ensemble learning (EL) algorithm

The implementation of the ensemble learning (EL) algorithm in fault detection and classification offers several advantages, such as reducing the need for a large training dataset and minimizing the risk of overfitting that is commonly encountered in single-classifier approaches. In this study, three algorithms are utilized namely Support Vector Machine (SVM) for its ability to tackle complex classification problems, Naïve Bayes (NB) for its effectiveness in handling small training datasets, and Logistic Regression (LR) for its linear classification capabilities, fast response time, and simplicity. In an EL model, each individual algorithm generates its own prediction, and the final prediction is determined by combining the results of each single classifier. To combine the results yielded by each learning algorithm, two widely accepted voting-based combination schemes can be applied: 1) majority voting scheme in which, as the name suggests, the classes are predicted according to the maximum number of votes, and 2) weighted voting scheme where

weights are allocated to the output predicted classes of each learning algorithm. In this study, a weighted voting scheme is used which computes the output prediction according to Eq. (18):

$$y' = \underset{j}{\operatorname{argmax}} \sum_{i=1}^m w_i p_{ji}(x) \quad (18)$$

where y' signifies the prediction output of EL model, j denotes the number of class labels, m is the number of learning algorithms, p_{ji} represents the prediction output of i_{th} learning algorithm based on probability occurrence, w_i shows the weight allocated to the i_{th} learning algorithm, i.e., $w = (w_1, w_2, w_3)$, $w_i \in [0, 1]$.

In order to assign the optimal weights to the predicted output of each classifier, a GA is used in this study. GA algorithms, belonging to the larger class of evolutionary algorithms (EA), suggest high-quality solutions to optimization problems based on biologically inspired operators, such as mutation, crossover and selection. To be precise, the loop starts with the algorithm generating a random set of chromosomes (also known as the initial population) (Kramer, 2017). In this study, each chromosome includes an array of real numbers between 0 and 1 that shows the set of assigned weights to each learning algorithm. Then, a fitness function measures the quality of the solution by evaluating the accuracy of the EL model with the resulting weights. Subsequently, the previously mentioned operators (mutation and crossover) take action and a new generation based on the specific mutation and crossover rates which determine the proportion of mutated and crossed-over chromosomes is created out of the most highly accurate ones. Finally, the loop ends when the termination condition is met.

3.4. Evaluation metrics

To validate the proposed model, the original dataset is divided into training (80 %) and validation (20 %) datasets. Also, above all, an unseen dataset which is not used during the processes of training and validation will be utilized to test the final multilayer model. In this study, two methods are presented to evaluate the learning algorithms in order to validate the proposed model. The first method is cross-validation (CV) which is used in feature selection as well as the training process. Among several types of CV techniques, such as hold-out, K -fold, stratified K -fold, and leave-one-out CV, the K -fold technique is used in this study which is the most common technique in cross-validation. In this technique, the whole training dataset is divided into K subsets which include equal number of data samples. Then, in each stage, one subset is set aside for the process of validation and other $K-1$ subsets are utilized to train the model. This iteratively continues until all subsets play their role once in the process of validation. In each stage, the validation subset yields an individual accuracy. The final accuracy is therefore the average of K individual accuracies obtained from the validation subset in each stage and is usually accompanied by another statistical metric which is the standard deviation. Although the value for K is not systematically formulated, it greatly depends on the size of the dataset. In this study, $K = 10$ is selected which is a normal number of folds in literature to initiate the process. The second method is a confusion matrix. A simple 3×3 confusion matrix is illustrated in Table 4. In the matrix shown in Table 4, the elements are formatted as “Predicted-Actual-True/False” in which the first number indicates the output predicted class by the algorithm, the second shows the actual

class the sample originally belongs to and the last boolean parameter verifies if the prediction is in accordance with the actual class. So, for instance, the term “1-1-T” signifies a sample which is correctly predicted as a member of the first class while in contrary “2-1-F” refers to a sample that the algorithm has categorized as the second class whereas it actually belongs to the first. Therefore the boolean variable has returned F standing for False.

To determine the practicability of a classification algorithm, “Accuracy” is a widespread metric which is measured by deploying the elements in a confusion matrix and is defined as Eq. (19):

$$\text{Accuracy} = \frac{\sum \text{correctly predicted samples}}{\sum \text{samples}} \quad (19)$$

It can be easily noticed that the numerator in Eq. (19) includes the sum of the diagonal elements in a confusion matrix. Also, the denominator comprises the number of all samples regardless of whether they are correctly or incorrectly predicted which, in fact, is the sum of all the elements in a confusion matrix.

However, “Accuracy” is not the only metric used in this study to assess the performance of the classifiers. Apart from that, two other important criteria namely “Recall” and “Precision” are calculated to evaluate the performance of the algorithms. Both criteria present the proportion of the number of correctly predicted samples, the former to the number of samples actually belong to the under-investigation class and the latter to the number of samples which are predictably categorized as a member of the certain class the criteria are being measured for. Eq. (20) and (21) mathematically formulates “Recall” and “Precision” for the i_{th} class, respectively:

$$\text{Recall} = \frac{(i - i - T)}{(i - i - T) + \sum_{j \neq i} (j - i - F)} \quad (20)$$

$$\text{Precision} = \frac{(i - i - T)}{(i - i - T) + \sum_{j \neq i} (i - j - F)} \quad (21)$$

where in both equations, the numerator, as previously mentioned, denotes the number of correctly predicted data samples related to the i_{th} class and j in the denominator is the number of the class unequal to i in which the boolean parameter is always false. Simply, the denominator in “Recall” and “Precision”, respectively sum up all the elements in the i_{th} row and column in the confusion matrix presented in Table 4.

4. Simulation results

In order to evaluate the performance of the proposed multilayer method (see Fig. 7), initially, a simulation is arranged in MATLAB/Simulink. In this study, two scenarios are considered, one with and one without the feature selection algorithm to further analyze and compare the results.

4.1. Data acquisition process

In this study, 3170 training as well as 456 unseen data samples including LL, LG, OC, array degradation, and string degradation faults as well as normal conditions are obtained from the simulation process and

Table 4
A simple 3×3 confusion matrix.

		Predicted Classes		
		Class 1	Class 2	Class3
Actual Classes	Class 1	1-1-T	2-1-F	3-1-F
	Class 2	1-2-F	2-2-T	3-2-F
	Class 3	1-3-F	2-3-F	3-3-T

Table 5
The obtained data from the simulation process.

Fault type	Simulation process	
	Number of original data	Number of unseen data
LL faults	570	120
LG faults	570	120
OC faults	433	60
Array degradation faults	582	48
String degradation faults	582	48
Normal conditions	433	60

summarized in detail in Table 5. Note that unseen data samples are used only in the process of testing the fully trained and validated model, are completely new to final model, and have not been utilized at all in the processes of training and validation of the model. Also, Table 6 provides statistical information on sixteen extracted features from the simulation training dataset, both before and after data normalization. The information includes Mean, Median, Mode, Standard deviation (Std) as well as the minimum (Min) and maximum (Max) values of each feature (especially before normalization) which are utilized in min-max normalization technique.

It is important to note that the simulated faults and normal conditions are subjected to various conditions, including different environmental conditions (i.e., different irradiance and temperature values), different fault impedance values, and various mismatch levels. To create a dataset under normal conditions, different combinations of temperature and irradiation are applied. The combinations are similarly used in OC faults except that in OC faults only two out of three strings are involved in the process of simulation. The temperature and irradiation used in this process range 0–40 °C and 200–1000 W/m², respectively. In degradation faults, a third parameter is also added which is the emulated series impedance values representing the amount of degradation. The impedance values are considered between 2 and 15 Ω with 3-ohm increasing increments for creating the training dataset while in the process of testing via unseen dataset, the impedance values are set to 4, 8 and 12 Ω. As for the LL and LG faults, mismatch levels and fault impedance values are two key parameters which the simulation is based on. Table 7 details different values of these two parameters applied in this study.

4.2. The performance of the multilayer model

4.2.1. The first layer

The first layer in the proposed multilayer model is named the fault-current-based splitter (FCBS) and acts as a binary categorizer that classifies the data samples into two main categories: 1) zero- to low-current faults which include normal conditions as well as OC and string degradation faults and 2) high-current faults, such as array degradation faults along with LL and LG faults under several mismatch levels and fault impedances (see Fig. 7). The number of data samples used in the processes of training, validation and testing is listed in detail in Table 8. As presented in Table 8, in the process of training, 1173 and 1363 samples are considered which are labeled as first and second class, respectively. During the validation process, the number of samples used in the first and second classes is turned into 275 and 359, respectively. Finally, Table 8 shows that 168 unseen data samples labeled as the first class and 288 as the second class are input to the model in the test process. In

Table 7

Different values for two main parameters in LL and LG faults.

Simulation process	
Parameter	Different values
Mismatch percentages (%)	10, 20, 30, 40, 50
Fault impedance values (Ω)	0, 5, 10, 15, 20, 25

addition, Table 9 illustrates the results of the proposed model after the training process. Firstly, it shows the most effective features selected by SFFS. The features are presented in Table 9 according to their assigned rank, reflecting the effectiveness of their selection by SFFS. Therefore, in the first layer, the selected features are $f_3, f_{11}, f_2, f_9, f_5$ and f_{14} . Moreover, Table 9 lists the optimized weights using a GA for each individual classifier (i.e., SVM, NB and LR). In this case, the optimized weights in the first layer for LR, NB and SVM are 0.21, 0.11 and 0.67, respectively. Finally, in terms of the accuracy, Table 9 displays that the model is evaluated by training accuracy which more precisely refers to the mean accuracy of the 10-fold CV process as well as the standard deviation. Therefore, it is evident that in the first layer, the accuracy of the proposed model is 96.37 % and the standard deviation is 1.5 %. The model is also compared in all the aforementioned items to individual LR, SVM and NB as well as a majority voting-based EL model and clearly demonstrates the highest accuracy along with the lowest standard deviation in the first layer. Also, in Table 10 which shows the process of validation, firstly the confusion matrices of the proposed model in both scenarios (S_1 and S_2) are presented. It is worth reminding that S_1 and S_2 refer to two scenarios excluding and including the SFFS, respectively. The accuracy column shows that the proposed method owns the accuracy of 94.63 % and 96.05 % during the first and second scenario, respectively. The 1.42 % increase in the accuracy in the second scenario in comparison to the first scenario during the validation process confirms the effectiveness of the SFFS. Finally, two other metrics are presented in Table 10 with the abbreviation P_i and R_i which stand for “Precision” and “Recall”, respectively while i refers to the number of the class. Therefore, for instance, P_1 shows the Precision of the first class while R_2 displays the Recall for the second class. Accordingly, Table 10 depicts that in the first layer during the first scenario the Precisions are 89 % and 100 % and the Recalls are 100 % and 91 % for the first and the second class, respectively. Furthermore, in the second scenario where the SFFS algorithm intervenes, the Precisions turn into 92 % and 100 % and the Recalls are 100 % and 93 % for the first and the second class, respectively. According to Table 10, the proposed model is also compared to the majority voting-based EL model in the process of validation and shows a better performance especially in the second scenario. Eventually, the results of the test process in the first layer can be observed in

Table 6

Statistical specifications of the training dataset used in the simulation stage.

	Before Normalization						After Normalization					
	Min	Max	Mode	Median	Mean	Std	Min	Max	Mode	Median	Mean	Std
f_1	0.132057	1.016421	0.496842	0.571384	0.603429	0.253901	0	1	0.412483	0.496771	0.533007	0.2871
f_2	0.1077	1.0175	0.993	0.9847	0.892548	0.200377	0	1	0.973071	0.963948	0.86266	0.220243
f_3	0.105283	1.10717	1.051817	0.97503	0.853199	0.24769	0	1	0.944752	0.86811	0.746508	0.247224
f_4	0.132552	1.015482	0.799387	0.552638	0.576874	0.233563	0	1	0.755252	0.475786	0.503236	0.264531
f_5	0.128063	0.991677	0.976842	0.535544	0.56735	0.234832	0	1	0.982822	0.471833	0.508662	0.271918
f_6	0.101109	0.963481	0.925287	0.901642	0.795682	0.208529	0	1	0.955711	0.928292	0.805421	0.241809
f_7	0.119722	9.6453	1.081577	0.684801	0.829653	0.785374	0	1	0.100976	0.059322	0.074529	0.082449
f_8	0.628836	1.111728	1.005967	0.992529	0.942425	0.131332	0	1	0.780984	0.753157	0.649398	0.27197
f_9	0.151417	1.032468	0.999556	0.994543	0.967344	0.071008	0	1	0.962644	0.956955	0.926084	0.080595
f_{10}	−0.07041	−0.00039	−0.00446	−0.0026	−0.00488	0.007004	0	1	0.941936	0.968543	0.935964	0.100037
f_{11}	−0.71594	−0.0202	−0.14475	−0.07966	−0.09023	0.066473	0	1	0.820984	0.914536	0.899351	0.095542
f_{12}	−0.06933	−0.00026	−0.00257	−0.00146	−0.00324	0.006548	0	1	0.995288	0.982615	0.956832	0.094805
f_{13}	−0.15682	−0.00057	−0.00846	−0.0047	−0.01217	0.018173	0	1	0.94949	0.973533	0.925734	0.116306
f_{14}	−0.32758	−0.0039	−0.0352	−0.02141	−0.02721	0.02709	0	1	0.903309	0.945913	0.927995	0.083694
f_{15}	−0.80574	−0.02971	−0.19262	−0.11879	−0.12463	0.082366	0	1	0.790066	0.885201	0.87768	0.106138
f_{16}	0.146387	1.11693	1.041303	0.985072	0.918081	0.169027	0	1	0.922078	0.864141	0.795116	0.174157

Table 8

Number of data samples used in training, validation and test processes.

Layer	Train dataset			Validation dataset			Test (Unseen) dataset			Event Rate (%)
	First class	Second class	Third class	First class	Second class	Third class	First class	Second class	Third class	
1	1173	1363		275	359		168	288		1173/2536 = 46.25 %
2	339	819		94	196		60	108		339/1158 = 29.27 %
3	356	456		77	126		60	48		356/812 = 43.84 %
4	445	458	474	125	112	108	120	120	48	445/1377 = 32.31 %
5	91	95	270	23	19	72	24	24	72	91/456 = 19.95 %
6	88	94	274	26	20	68	24	24	72	88/456 = 19.29 %

Table 9

The performance of the proposed model in training process.

Layer	Selected Features					Selected Weights			Train Accuracy (%) (Standard Deviation (%))				
	LR	NB	SVM	Majority voting-based EL	The proposed method	LR	NB	SVM	LR	NB	SVM	Majority voting-based EL	The proposed method (running time (s))
1	f_3, f_{15}, f_8	f_3, f_{15}	$f_3, f_{11}, f_2, f_8, f_{14}$	f_3, f_{15}, f_8	$f_3, f_{11}, f_2, f_9, f_5, f_{14}$	0.21	0.11	0.67	93.62 % (6.9 %)	91.34 % (8.4 %)	93.97 % (7.2 %)	93.87 % (7.2 %)	96.37 % (1.5 %) (44.6)
2	$f_{15}, f_{11}, f_6, f_{16}, f_8, f_2, f_4, f_5, f_7$	f_{15}, f_{16}, f_2, f_8	f_{15}, f_8, f_6, f_{16}	$f_{15}, f_{16}, f_6, f_2, f_8$	$f_{15}, f_{12}, f_2, f_3, f_8$	0.062	0.218	0.687	97.1 % (1.09 %)	88.68 % (1.3 %)	98.01 % (1.09 %)	95.59 % (1.1 %)	99.9 % (0.2 %) (20.08)
3	$f_{15}, f_{14}, f_8, f_{13}, f_6, f_{10}, f_9, f_4$	f_{15}, f_9	f_{15}, f_{14}	$f_{15}, f_{14}, f_8, f_9, f_2, f_6, f_{10}$	$f_{15}, f_{11}, f_{14}, f_2$	0.062	0.062	0.63	98.76 % (1.1 %)	80.17 % (2.8 %)	99 % (0.7 %)	98.76 % (1.2 %)	100 % (0 %) (18.08)
4	$f_2, f_9, f_{15}, f_{11}, f_{12}, f_{14}, f_{16}, f_5$	f_2, f_{16}, f_{15}, f_8	$f_2, f_{11}, f_{15}, f_9, f_{14}$	$f_2, f_{15}, f_9, f_{14}, f_{11}, f_{16}, f_{12}$	$f_2, f_{16}, f_8, f_{13}, f_9, f_{15}, f_4$	0.32	0.114	0.583	94.61 % (1.7 %)	91.2 % (1.9 %)	96.14 % (1.9 %)	93.43 % (7 %)	97.56 % (1.4 %) (22.65)
5	$f_6, f_{16}, f_{10}, f_{11}, f_4$	$f_6, f_{16}, f_{11}, f_{14}, f_4, f_8$	f_6, f_{16}, f_2	$f_6, f_{16}, f_{13}, f_3, f_2$	f_6, f_{16}, f_2	0.114	0.01	0.791	95.83 % (2 %)	95.16 % (3.9 %)	99.77 % (0.66 %)	98.89 % (1.1 %)	99.77 % (0.6 %) (14.67)
6	f_3, f_9, f_8, f_4	$f_3, f_{10}, f_5, f_{16}, f_6$	f_3, f_9, f_2	$f_3, f_{12}, f_{15}, f_9, f_2, f_8$	f_3, f_9, f_8, f_2	0.114	0.322	0.531	99.56 % (0.8 %)	92.95 % (3.5 %)	99.77 % (0.66 %)	99.56 % (0.8 %)	99.77 % (0.6 %) (13.54)

Table 11 which shows 95.17 % accuracy of the proposed model in the second scenario which is 1.48 % more than 93.69 % accuracy during the first scenario. The results are then compared to the majority voting-based EL model and prove to be more accurate especially during the second scenario where the majority voting-based EL model reaches a high accuracy of 93.2 % compared to the accuracy of the proposed model in the second scenario which is 95.17 %. Also, Fig. 10(a) and 10 (b) show the effectiveness of the SFFS algorithm and the performance of WEL algorithm during the feature selection process. The WEL algorithm performs the best and reaches the maximum accuracy using six features along with a very small amount of standard deviation (1.5 % according to Table 9, also see Fig. 10(b)) while this number is three for the majority voting-based EL algorithm which in contrary is accompanied by a considerably high standard deviation (7.2 % according to Table 9, also see Fig. 10(a)).

4.2.2. The second layer

After the “No” events are distinguished and categorized by the first layer, they should be classified into three different categories namely normal conditions, OC faults and array degradation faults. To this end, two layers are provided such that one distinguishes between normal and low-current fault conditions (the second layer) and the other classifies the conditions which are recognized as the low-current faults (the third layer). Hence, the second layer consists of two classes namely normal

(first class) and low-current fault conditions (second class) (see Fig. 7). The number of data used for two mentioned classes in the second layer is respectively 339 and 819 during the process of training, 94 and 196 in the validation process and 60 and 108 in the testing stage and is shown in detail in Table 8. Furthermore, according to Table 9, the proposed model in the second layer during the process of training has obtained the accuracy of 99.9 % along with a 0.2 % of standard deviation with only five selected features which are the highest accuracy and lowest standard deviation in comparison to all the algorithms to which the model is compared. After the validation process (see Table 10), the model during the first and second scenarios present a 98.27 % and a 100 % of accuracy, respectively which are followed by almost similar results (98.8 % in the first and 100 % in the second scenario) during the process of testing the algorithm via unseen dataset (see Table 11). Also, Fig. 11 (a) and 11(b) depict the effectiveness of the SFFS algorithm and the performance of WEL algorithm during the process of feature selection. As shown in Fig. 11(a) and 11(b), both the proposed WEL algorithm and the majority voting-based EL algorithm reach their maximum accuracy using five features whereas the metric of standard deviation is again lower in the proposed method (0.2 % according to Table 9, also see Fig. 11(b)) in comparison to the majority voting-based EL algorithm (1.1 % according to Table 9, also see Fig. 11(a)).

Table 10

Detailed information of validation process including both scenarios.

Layer	Scenarios	Confusion Matrix		Accuracy (%)		Precision (%)		Recall (%)	
		Majority voting-based EL model	The proposed method	Majority voting-based EL model	The proposed method	Majority voting-based EL model	The proposed method	Majority voting-based EL model	The proposed method
1	S_1	$\begin{bmatrix} 274 & 1 \\ 33 & 326 \end{bmatrix}$	$\begin{bmatrix} 274 & 1 \\ 32 & 326 \end{bmatrix}$	94.63 %	94.63 %	$P1 = 89\%$ $P2 = 100\%$	$P1 = 89\%$ $P2 = 100\%$	$R1 = 100\%$ $R2 = 91\%$	$R1 = 100\%$ $R2 = 91\%$
	S_2	$\begin{bmatrix} 275 & 0 \\ 36 & 323 \end{bmatrix}$	$\begin{bmatrix} 275 & 0 \\ 25 & 334 \end{bmatrix}$	94.32 %	96.05 %	$P1 = 88\%$ $P2 = 100\%$	$P1 = 92\%$ $P2 = 100\%$	$R1 = 100\%$ $R2 = 90\%$	$R1 = 100\%$ $R2 = 93\%$
2	S_1	$\begin{bmatrix} 94 & 0 \\ 13 & 183 \end{bmatrix}$	$\begin{bmatrix} 91 & 3 \\ 2 & 194 \end{bmatrix}$	95.51 %	98.27 %	$P1 = 88\%$ $P2 = 100\%$	$P1 = 98\%$ $P2 = 98\%$	$R1 = 100\%$ $R2 = 93\%$	$R1 = 97\%$ $R2 = 99\%$
	S_2	$\begin{bmatrix} 88 & 6 \\ 8 & 188 \end{bmatrix}$	$\begin{bmatrix} 94 & 0 \\ 0 & 196 \end{bmatrix}$	95.17 %	100 %	$P1 = 92\%$ $P2 = 97\%$	$P1 = 100\%$ $P2 = 100\%$	$R1 = 94\%$ $R2 = 96\%$	$R1 = 100\%$ $R2 = 100\%$
3	S_1	$\begin{bmatrix} 77 & 0 \\ 6 & 120 \end{bmatrix}$	$\begin{bmatrix} 77 & 0 \\ 4 & 122 \end{bmatrix}$	97.04 %	98.02 %	$P1 = 93\%$ $P2 = 100\%$	$P1 = 95\%$ $P2 = 100\%$	$R1 = 100\%$ $R2 = 95\%$	$R1 = 100\%$ $R2 = 97\%$
	S_2	$\begin{bmatrix} 77 & 0 \\ 4 & 122 \end{bmatrix}$	$\begin{bmatrix} 77 & 0 \\ 1 & 125 \end{bmatrix}$	98.02 %	99.5 %	$P1 = 95\%$ $P2 = 100\%$	$P1 = 99\%$ $P2 = 100\%$	$R1 = 100\%$ $R2 = 97\%$	$R1 = 100\%$ $R2 = 99\%$
4	S_1	$\begin{bmatrix} 114 & 2 & 9 \\ 10 & 102 & 0 \\ 0 & 0 & 108 \end{bmatrix}$	$\begin{bmatrix} 119 & 2 & 4 \\ 8 & 104 & 0 \\ 2 & 0 & 106 \end{bmatrix}$	93.91 %	95.36 %	$P1 = 92\%$ $P2 = 98\%$ $P3 = 92\%$	$P1 = 92\%$ $P2 = 98\%$ $P3 = 96\%$	$R1 = 91\%$ $R2 = 91\%$ $R3 = 100\%$	$R1 = 95\%$ $R2 = 93\%$ $R3 = 98\%$
	S_2	$\begin{bmatrix} 119 & 0 & 6 \\ 10 & 102 & 0 \\ 0 & 0 & 108 \end{bmatrix}$	$\begin{bmatrix} 123 & 2 & 0 \\ 8 & 104 & 0 \\ 0 & 0 & 108 \end{bmatrix}$	95.36 %	97.1 %	$P1 = 92\%$ $P2 = 100\%$ $P3 = 95\%$	$P1 = 94\%$ $P2 = 98\%$ $P3 = 100\%$	$R1 = 95\%$ $R2 = 91\%$ $R3 = 100\%$	$R1 = 98\%$ $R2 = 93\%$ $R3 = 100\%$
5	S_1	$\begin{bmatrix} 23 & 0 & 0 \\ 2 & 17 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	$\begin{bmatrix} 23 & 0 & 0 \\ 0 & 19 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	98.24 %	100 %	$P1 = 92\%$ $P2 = 100\%$ $P3 = 100\%$	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$R1 = 100\%$ $R2 = 89\%$ $R3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$
	S_2	$\begin{bmatrix} 23 & 0 & 0 \\ 1 & 18 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	$\begin{bmatrix} 23 & 0 & 0 \\ 0 & 19 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	99.12 %	100 %	$P1 = 96\%$ $P2 = 100\%$ $P3 = 100\%$	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$R1 = 100\%$ $R2 = 95\%$ $R3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$
6	S_1	$\begin{bmatrix} 26 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 68 \end{bmatrix}$	$\begin{bmatrix} 26 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 68 \end{bmatrix}$	100 %	100 %	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$
	S_2	$\begin{bmatrix} 26 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 68 \end{bmatrix}$	$\begin{bmatrix} 26 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 68 \end{bmatrix}$	100 %	100 %	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$

Table 11

Detailed information of test process including both scenarios.

Layer	Scenarios	Confusion Matrix		Accuracy (%)		Precision (%)		Recall (%)	
		Majority voting-based EL model	The proposed method	Majority voting-based EL model	The proposed method	Majority voting-based EL model	The proposed method	Majority voting-based EL model	The proposed method
1	S_1	$\begin{bmatrix} 167 & 1 \\ 30 & 258 \end{bmatrix}$	$\begin{bmatrix} 167 & 1 \\ 30 & 258 \end{bmatrix}$	93.2 %	93.69 %	$P1 = 85\%$ $P2 = 100\%$	$P1 = 85\%$ $P2 = 100\%$	$R1 = 99\%$ $R2 = 90\%$	$R1 = 99\%$ $R2 = 90\%$
	S_2	$\begin{bmatrix} 168 & 0 \\ 31 & 257 \end{bmatrix}$	$\begin{bmatrix} 168 & 0 \\ 22 & 266 \end{bmatrix}$	93.2 %	95.17 %	$P1 = 84\%$ $P2 = 100\%$	$P1 = 88\%$ $P2 = 100\%$	$R1 = 100\%$ $R2 = 89\%$	$R1 = 100\%$ $R2 = 92\%$
2	S_1	$\begin{bmatrix} 58 & 2 \\ 0 & 108 \end{bmatrix}$	$\begin{bmatrix} 58 & 2 \\ 0 & 108 \end{bmatrix}$	98.8 %	98.8 %	$P1 = 100\%$ $P2 = 98\%$	$P1 = 100\%$ $P2 = 98\%$	$R1 = 97\%$ $R2 = 100\%$	$R1 = 97\%$ $R2 = 100\%$
	S_2	$\begin{bmatrix} 56 & 4 \\ 0 & 108 \end{bmatrix}$	$\begin{bmatrix} 60 & 0 \\ 0 & 108 \end{bmatrix}$	97.61 %	100 %	$P1 = 100\%$ $P2 = 96\%$	$P1 = 100\%$ $P2 = 100\%$	$R1 = 93\%$ $R2 = 100\%$	$R1 = 100\%$ $R2 = 100\%$
3	S_1	$\begin{bmatrix} 60 & 0 \\ 1 & 47 \end{bmatrix}$	$\begin{bmatrix} 60 & 0 \\ 0 & 48 \end{bmatrix}$	99.07 %	100 %	$P1 = 98\%$ $P2 = 100\%$	$P1 = 100\%$ $P2 = 100\%$	$R1 = 100\%$ $R2 = 98\%$	$R1 = 100\%$ $R2 = 100\%$
	S_2	$\begin{bmatrix} 60 & 0 \\ 0 & 48 \end{bmatrix}$	$\begin{bmatrix} 60 & 0 \\ 0 & 48 \end{bmatrix}$	100 %	100 %	$P1 = 100\%$ $P2 = 100\%$	$P1 = 100\%$ $P2 = 100\%$	$R1 = 100\%$ $R2 = 100\%$	$R1 = 100\%$ $R2 = 100\%$
4	S_1	$\begin{bmatrix} 101 & 0 & 19 \\ 0 & 120 & 0 \\ 0 & 0 & 48 \end{bmatrix}$	$\begin{bmatrix} 112 & 0 & 8 \\ 0 & 120 & 0 \\ 0 & 0 & 48 \end{bmatrix}$	93.4 %	97.22 %	$P1 = 100\%$ $P2 = 100\%$ $P3 = 72\%$	$P1 = 100\%$ $P2 = 100\%$ $P3 = 86\%$	$R1 = 84\%$ $R2 = 100\%$ $R3 = 100\%$	$R1 = 93\%$ $R2 = 100\%$ $R3 = 100\%$
	S_2	$\begin{bmatrix} 100 & 2 & 18 \\ 0 & 120 & 0 \\ 0 & 0 & 48 \end{bmatrix}$	$\begin{bmatrix} 120 & 0 & 0 \\ 0 & 120 & 0 \\ 0 & 0 & 48 \end{bmatrix}$	93.05 %	100 %	$P1 = 100\%$ $P2 = 98\%$ $P3 = 73\%$	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$R1 = 83\%$ $R2 = 100\%$ $R3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$
5	S_1	$\begin{bmatrix} 24 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	$\begin{bmatrix} 24 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	100 %	100 %	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$
	S_2	$\begin{bmatrix} 24 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	$\begin{bmatrix} 24 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	100 %	100 %	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$
6	S_1	$\begin{bmatrix} 24 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	$\begin{bmatrix} 24 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	100 %	100 %	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$
	S_2	$\begin{bmatrix} 24 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	$\begin{bmatrix} 24 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 72 \end{bmatrix}$	100 %	100 %	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$P1 = 100\%$ $P2 = 100\%$ $P3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$	$R1 = 100\%$ $R2 = 100\%$ $R3 = 100\%$

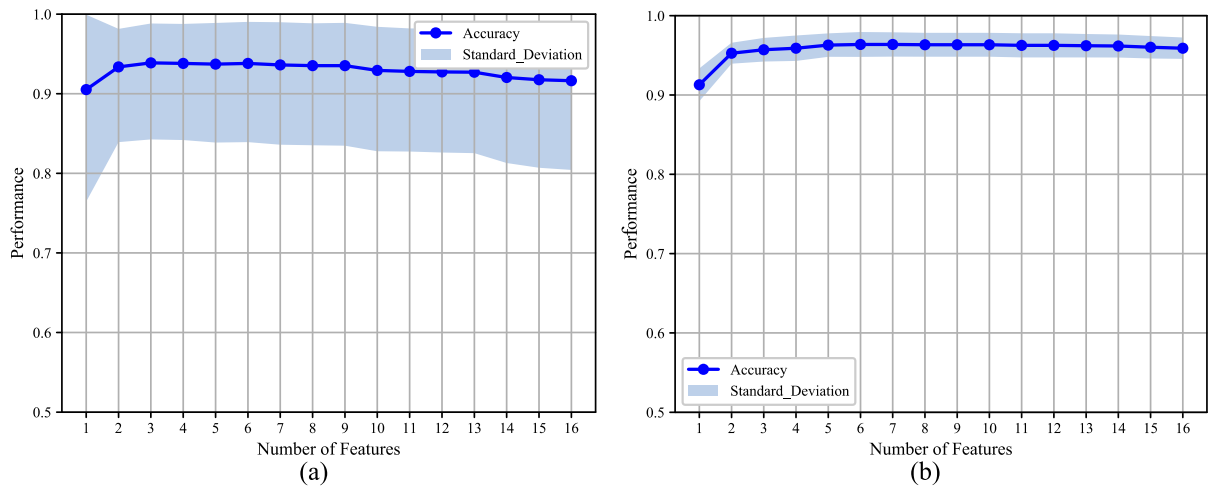


Fig. 10. The performance of models in the training process using the SFFS algorithm (a) majority voting-based EL algorithm in the first layer (b) the proposed WEL algorithm in the first layer.

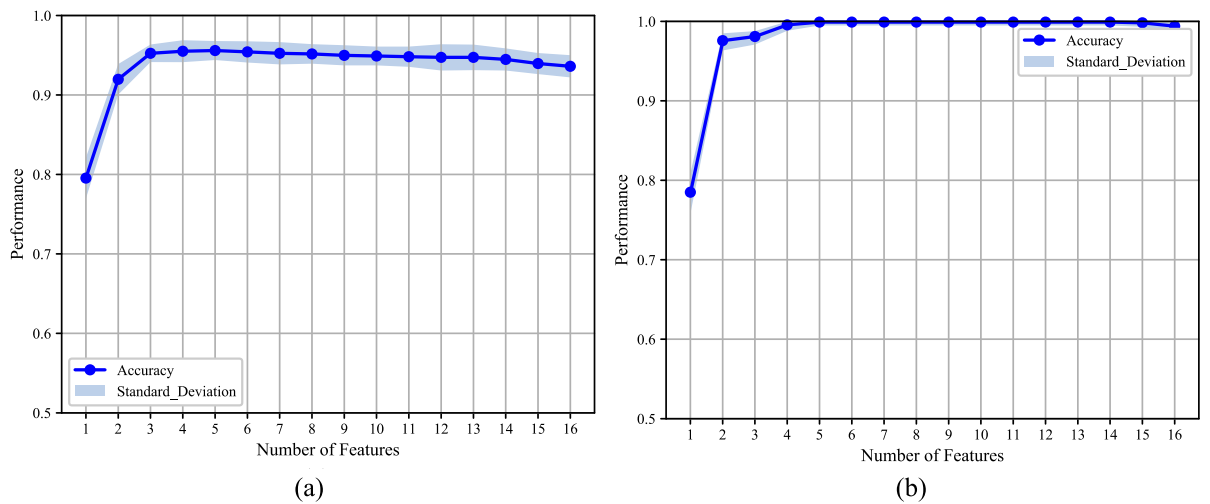


Fig. 11. The performance of models in the training process using the SFFS algorithm (a) majority voting-based EL algorithm in the second layer (b) the proposed WEL algorithm in the second layer.

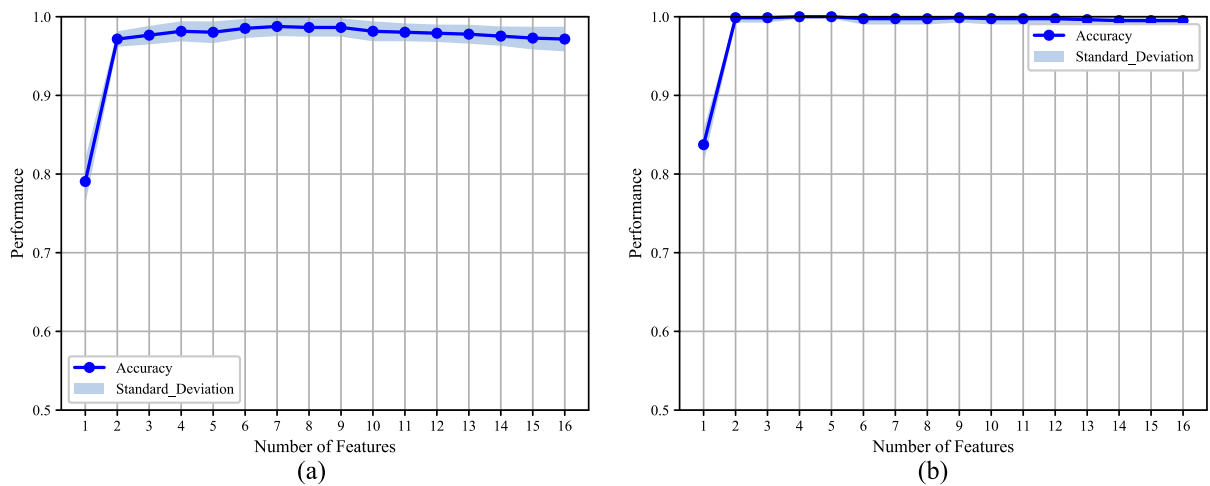


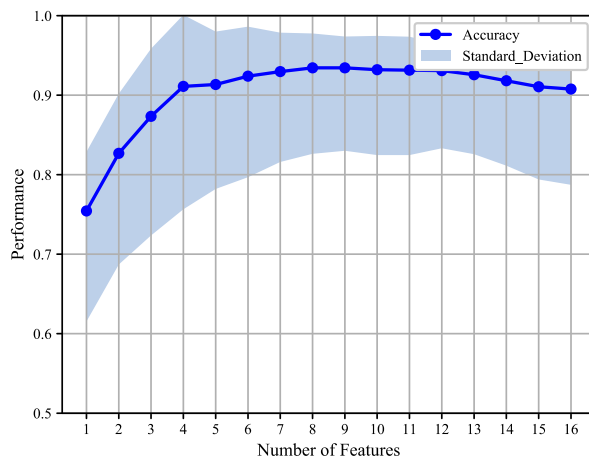
Fig. 12. The performance of models in the training process using the SFFS algorithm (a) majority voting-based EL algorithm in the third layer (b) the proposed WEL algorithm in the third layer.

4.2.3. The third layer

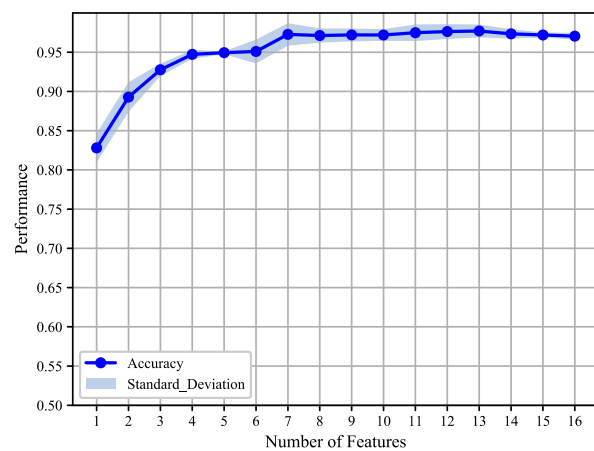
The third layer is responsible for classifying the events that have been identified as faulty events by the second layer. The faults investigated in this layer are OC and string degradation faults implying the existence of two classes in this layer attributed to the aforementioned faults as first and second class, respectively (see Fig. 7). The number of data used in the processes of training, validation and testing is listed in detail in Table 8. Besides, Table 9 depicts that after the training process, the proposed method in the third layer shows a remarkable performance with a 100 % of accuracy along with no standard deviation. Also, in the process of validation (see Table 10), the third layer has gained the accuracy of 98.02 % in the first and 99.5 % in the second scenario which are both higher than those of majority voting-based EL model. Then, the numbers grow to a high of 100 % when the third layer is tested via unseen dataset (see Table 11). Also, Fig. 12(a) and 12(b) demonstrate the usefulness of the SFFS algorithm and the performance of WEL algorithm in comparison to the majority voting-based EL algorithm during the process of feature selection. The figures show that the proposed method can perform the best in the third layer using four features along with 0 % of standard deviation while the number is seven for the majority voting-based EL algorithm and 1.2 % of standard deviation.

4.2.4. The fourth layer

In the first layer of the multilayer model, the data samples are labeled as either zero- to low-current faults or high-current faults. The former class (the “No” output in the first layer) is probed through the second and third layers and set the “Yes” output aside to be separately examined now in the fourth layer. The rest of the fault types which are LL, LG and array degradation faults are allotted to the fourth layer to be detected and categorized. Therefore, as expected, this layer contains three classes which are named as the first, second and third class after the above-mentioned faults, respectively (see Fig. 7). The number of data samples used in the processes of training, validation and testing is listed in detail in Table 8. Moreover, Table 9 shows that the fourth layer has resulted in the accuracy of 97.56 % and a 1.4 % of standard deviation during the process of training. The results of the validation process in the fourth layer are 95.36 % and 97.1 % accuracy in the first and second scenario, respectively (see Table 10). Eventually, the fourth layer is sent to a testing process via unseen data samples where it proves a 100 % of accuracy in the second scenario (see Table 11). Also, Fig. 13(a) and 13(b) show the effectiveness of the SFFS algorithm and the performance of WEL algorithm compared to the majority voting-based EL algorithm during the process of feature selection.



(a)



(b)

Fig. 13. The performance of models in the training process using the SFFS algorithm (a) majority voting-based EL algorithm in the fourth layer (b) the proposed WEL algorithm in the fourth layer.

4.2.5. The fifth layer

As previously explained, LL faults with low mismatch percentages are likely to remain undetected. To address this problem, the severity of LL faults is detected through the fifth layer, which considers three classes: 10 %, 20 %, and over 20 % mismatch level (first, second and third classes, respectively). The number of data samples used in the processes of training, validation and testing is presented in Table 8. According to Table 9, the process of training ends with the accuracy of 99.77 % and the lowest standard deviation of 0.6 %. It turns into a 100 % of accuracy in the process of validation in both scenarios (see Table 10). Finally, the process of testing the fifth layer via unseen dataset depicts a high accuracy of 99.16 % and 100 % in LL faults severity detection in the first and second scenario, respectively (see Table 11). Fig. 14(a) and Fig. 14(b) depict the effectiveness of the SFFS algorithm and the performance of WEL algorithm during the feature selection process. As shown in Fig. 14, compared to majority voting-based EL algorithm which require five features (see Fig. 14(a)), the proposed WEL algorithm performs the best using only three features (see Fig. 14(b)) which reflects the maximum accuracy and minimum standard deviation (see Table 9).

4.2.6. The sixth layer

In contrast to LL faults, LG faults are more likely to remain undetected under high mismatch levels. The final layer is responsible for classifying the severity of LG faults, taking into consideration three classes: 50 %, 40 %, and under 40 % mismatch level (first, second and third classes, respectively). The number of data samples used in the process of training, validation and testing is listed in detail in Table 8. Here, it should be noted that according to the event rate calculated using Eq. (22) in each layer (see Table 8), none of the layers exhibit an imbalanced state with respect to the number of data in their classes.

$$\text{Event rate} = \text{Number of minority class data} / \text{total dataset} \quad (22)$$

Also, the model demonstrates a 99.77 % accuracy along with 0.6 % standard deviation during the training process according to Table 9 and a notable 100 % accuracy in both scenarios during both validation and test processes according to Table 10 and Table 11. Moreover, Fig. 15(a) and 15(b) show the usefulness of the SFFS algorithm and the performance of WEL algorithm during the process of feature selection. It should be noted that the hyperparameters of each learning algorithm (i. e., SVM, LR, and NB) are optimized using the Grid-Search method. This method involves a comprehensive search of different combinations of the hyperparameters within a specified range. It is worth mentioning that Naive Bayes (NB) algorithm does not have any hyperparameters to

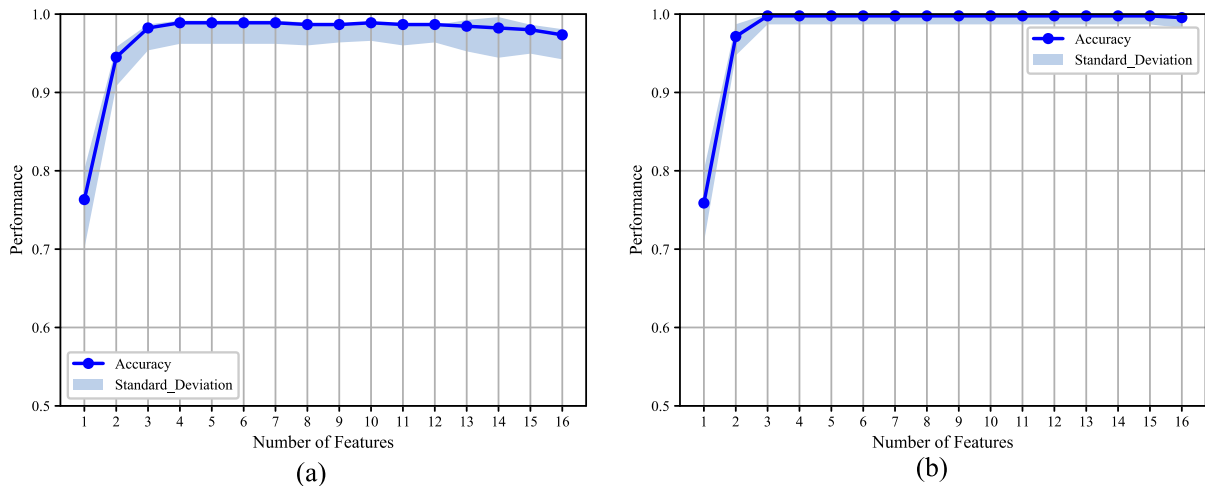


Fig. 14. The performance of models in the training process using the SFFS algorithm (a) majority voting-based EL algorithm in the fifth layer (b) the proposed WEL algorithm in the fifth layer.

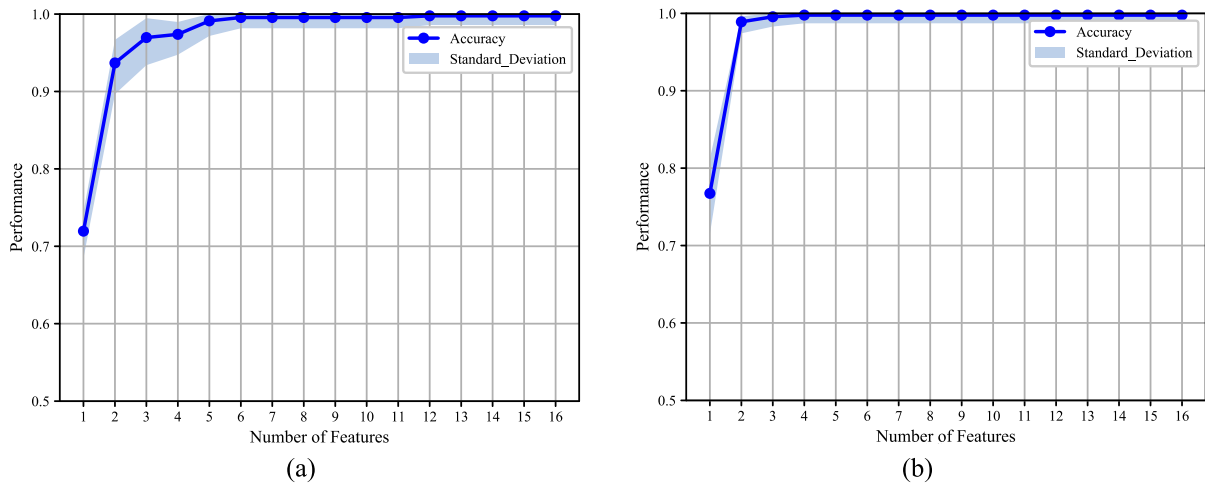


Fig. 15. The performance of models in the training process using the SFFS algorithm (a) majority voting-based EL algorithm in the sixth layer (b) the proposed WEL algorithm in the sixth layer.

be optimized. The optimized hyperparameters for the learning algorithms are presented in the Table 12. Also, the running time for the total evolution generation is recorded in the proposed model for each layer. As shown in Table 9, the execution time on a standard 4-core CPU varies based on the number of utilized datasets. In this study, a smaller number of datasets is needed for training in each layer, therefore, the overall execution time of the algorithm remains manageable.

4.3. Comparison of simulation results

The proposed multilayer model has been shown to be highly accurate

while it requires only a small training dataset to detect, classify and identify the severity of specific faults in PV systems. To further demonstrate the capability of the proposed multilayer model, two different comparisons are provided below.

1) Firstly, a uni-layer model which utilizes the same dataset is presented and compared to the majority voting-based EL algorithm. Accordingly, there is only a single layer which is trained to detect and classify the faulty and nomral conditions. Here, the dataset consists of six classes namely normal conditions along with LL, LG, OC, array and string degradation faults. It should be noted that in this model, the severity of faults will not be detected. Fig. 16(a) and 16(b) show the

Table 12
Optimal Parameters of the Learning Algorithms in this study.

Detail	Parameters	Layer					
		1	2	3	4	5	6
SVM	Cost C	1	5.1	2.4	68.1	10.2	8.1
	Kernel function	Radial Basis Function	Radial Basis Function	Radial Basis Function	Radial Basis Function	Radial Basis Function	Radial Basis Function
	γ in kernel function	0.01	0.6	0.8	10	0.2	0.4
LR	C	20.4	10	8.2	4.8	100	100

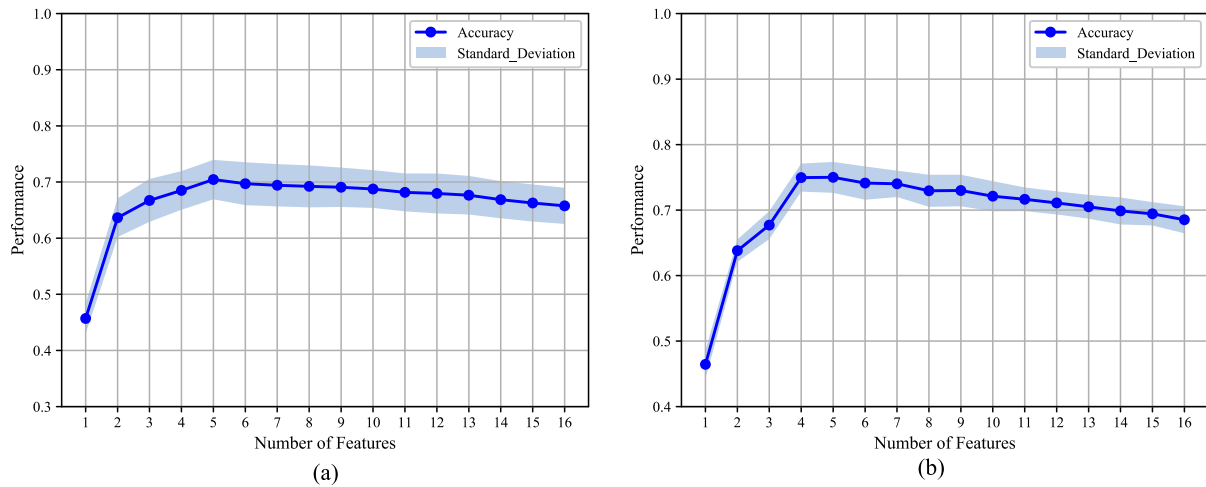


Fig. 16. The performance of models in the training process using the SFFS algorithm in the first condition (a) majority voting-based EL model (b) the uni-layer model.

validation results of the uni-layer model in comparison to the majority voting-based EL algorithm. As can be seen, the uni-layer model shows a higher accuracy than the majority voting-based EL algorithm. However, Fig. 16(b) which shows the performance of the uni-layer model, displays that the accuracy of the uni-layer model does not exceed 75 % which clearly indicates the superiority of the multilayer model in 1) reducing the data samples and 2) heightening the accuracy.

2) The second condition is similar to the first one except that here, the severity of LL and LG faults are also considered. Therefore, in this condition, ten classes namely normal conditions, OC faults, LL faults with 10 %, 20 % and over 20 % mismatch, LG faults with 50 %, 40 % and less than 40 % mismatch as well as array and string degradation faults are considered. Fig. 17(a) and 17(b) demonstrate the validation results of the uni-layer model in comparison to the majority voting-based EL algorithm. It is evident that firstly, the uni-layer model is shown to be more accurate than the majority voting-based EL model with 70 % of accuracy in comparison to maximum accuracy of 65 %. Subsequently, in comparison to Fig. 16, here, the figures prove that as the number of classes increases, the accuracy of fault detection and classification tends to reduce due to a reduction in the ratio of the number of data samples in each class to the total number of data samples.

5. Experimental Results

5.1. Experimental setup and dataset creation

In this section, a homemade I-V curve tracer is designed consisting of a DC-DC converter, a microcontroller to control the converter switch, voltage and current measurement sensors and the gate driver circuit (see Fig. 18). As shown in Fig. 18, the microcontroller performs based on an ARM microcontroller which controls the duty cycle to elicit the desired features from the I-V curve. The testing algorithm of the I-V curve is programmed using STM32f103C8T6 ARM. To provide further explanation, it should be mentioned that STM32F103C8T6 is a medium density performance line, ARM Cortex-M3 32bit microcontroller in 48-pin LQFP package. It incorporates high performance RISC core with 72 MHz operating frequency, high speed embedded memories, extensive range of enhanced I/Os and peripherals connected to two APB buses. During the test, the controller reads two input values which are the voltage and current of the PV array. The peak of the signal produced by the controller is 3.3 V while the switch Q requires at least 15 V to stay on. Therefore, a gate driver is applied to provide the sufficient gate voltage for the MOSFET. The controller and output circuits of the driver are supplied by two voltage regulators, one of which outputs 3.3 V for the ARM controller and the other produces a controllable 15-V voltage for

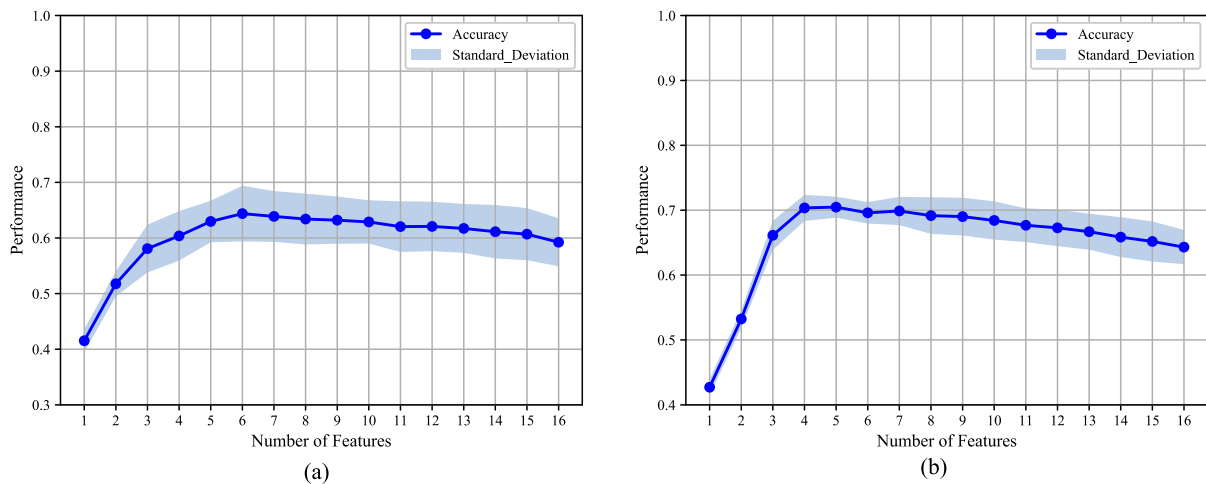


Fig. 17. The performance of models in the training process using the SFFS algorithm in the second condition (a) majority voting-based EL model (b) the uni-layer model.

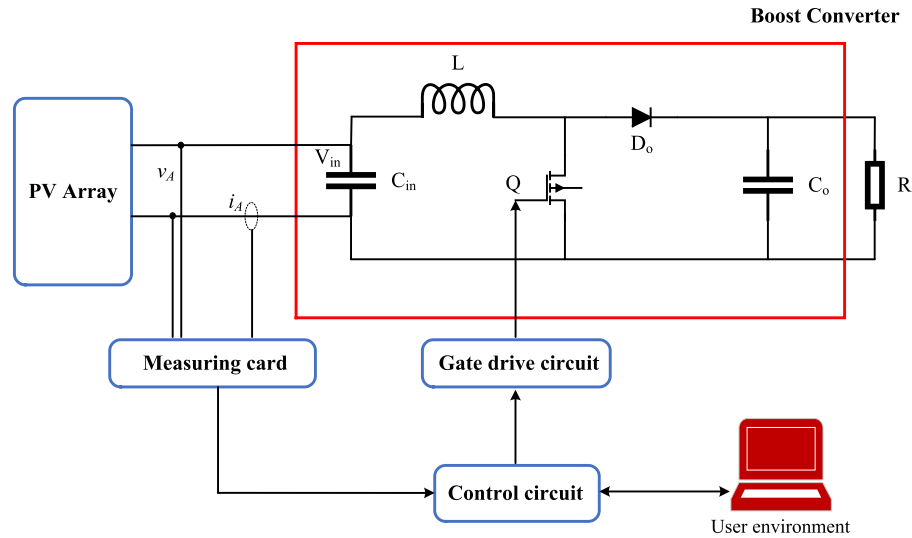


Fig. 18. Schematic of homemade I-V curve tracer.

the gate driver. The proposed tracking algorithm is so simple in a way that initially the PV array performs with a low-resistive load and a high amount of current. This load is parallel to the boost converter which is excited by a signal with a low duty cycle (approximately 10 %). Here, the PV array is expected to produce an amount of current nearly similar to the short circuit current when the magnitude of voltage is low. The measurement is repeated with an incremental increase in duty cycle until it reaches a specified range which is above 80 %. Finally, the open circuit voltage is separately calculated by measuring the terminal

voltage of the PV array under the open circuit condition.

In order to evaluate the designed homemade I-V curve tracer, a sample stand-alone PV system is assembled. In this experimental setup, a PV array is created which is comprised of three strings, each consists of six series Yingli YL010D-18b PV modules, see Fig. 19.

After the PV array setup, a small experimental sample of a boost converter and a data acquisition system is created to obtain the required data to plot the I-V characteristic curve. The system includes a DC/DC boost converter, an STM32F103C8T6 ARM microcontroller, a

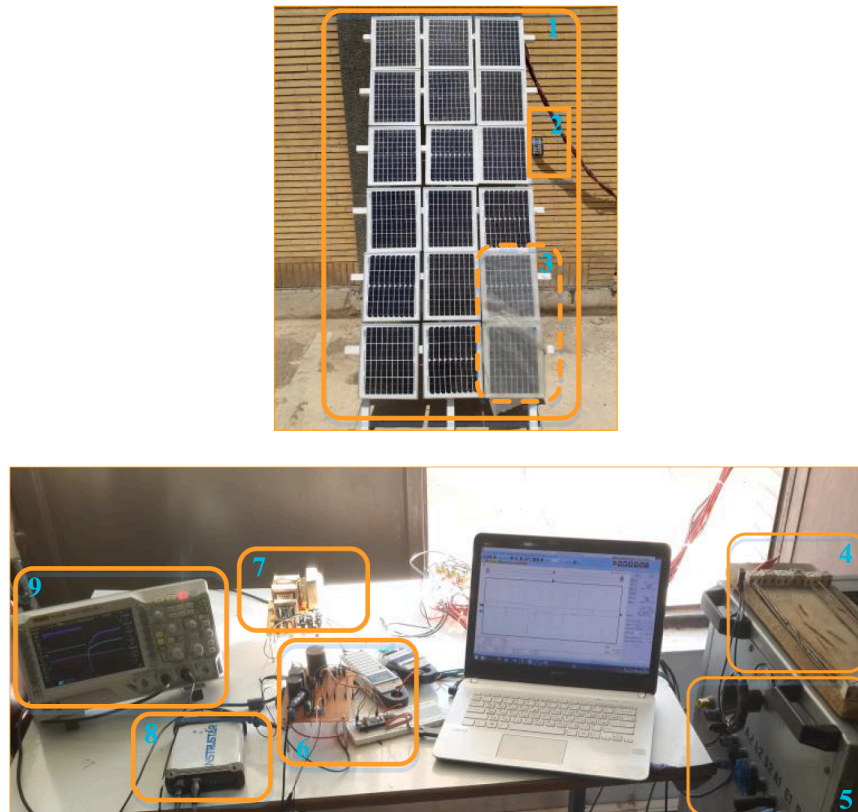


Fig. 19. Experimental setup: 1) PV array; 2) Solar analyzer; 3) Shading effect; 4) Fault impedance and switch; 5) R-Load; 6) DC/DC converter and ARM controller; 7) Gate driver circuit; 8) Data acquisition oscilloscope; 9) oscilloscope display.

measuring system, an InstruStarISDS205A oscilloscope card and a resistive load. It should be mentioned that, InstruStarISDS205A oscilloscope is a well-designed 20-meter bandwidth, 48 M sampling rate. It also features two channels, alternate support for X, Y, and XY dual channel virtual oscilloscope, spectrum analyzer, and data recorder alternating mode. This device can communicate with PC via high-speed USB2.

In this study, 1740 training data and 240 testing samples are obtained through the experimental process and are detailed in Table 13.

The above-mentioned faults are implemented under the same conditions as those in simulation process. The only difference is that here, the mismatch percentages and fault impedance values for LL and LG faults slightly differ from those in simulation process. Table 14 lists the values of these two parameters in the experimental process.

5.2. Experimental results

In this section, the performance of the proposed multilayer model is investigated approaching the second scenario where the SFFS algorithm is applied in each layer. To implement the method, initially, the WEL algorithm is trained in each individual layer via training dataset and sequently is tested using the test data samples. Table 15 displays the results and confirms that the proposed multilayer model can detect, classify and distinguish various faults and normal conditions with a very high accuracy. According to Table 15, in the first layer, the model reached the accuracy of 95.32 % in the process of training which is further improved to 96.25 % during the testing stage. The second and third layers demonstrate a remarkable performance with a 100 % of testing accuracy. In comparison, the fourth layer shows a training accuracy of 96.17 % and a testing of 97.33 %. Finally, the last two layers have again performed brilliantly and both show a maximum 100 % of accuracy in the process of testing via unseen data samples.

5.3. Comparison of experimental results and discussion

To further verify the superiority of the proposed multilayer model, it is now compared thoroughly in Table 16 to various most recent studies in the same area of research. In order to make a fair comparison, several factors have been considered: a) very recent studies (between 2022 and 2024) have been nominated which guarantees that the findings and results in studies are not old-fashioned b) studies including modern techniques, such as deep learning, etc. have been selected so that the proposed model is not compared with old fault detection techniques c) studies and models have been compared from various aspects and perspectives, such as the capability of severity identification which helps provide a more accurate protection, the number of required data samples to fully train the model which avoids model complexity, data acquisition challenges, data storing problems, etc., the consideration of critical fault detection conditions which make the process of fault detection extremely difficult, and the final experimental accuracies which shows how reliable the final model is to detect and classify various faults in PV arrays. According to the comparison made in Table 16, the following detailed discussion can be presented:

a) There are various models in literature, such as (Hajji et al., 2023),

Table 13

The obtained data from the experimental process.

Fault type	Experimental process	
	Number of training data	Number of testing data
LL faults	300	60
LG faults	300	60
OC faults	300	30
Array degradation faults	270	30
String degradation faults	270	30
Normal conditions	300	30

Table 14

Different values of mismatch percentage and fault impedance in LL and LG faults.

Experimental process	Different values	
	Parameter	
	LL faults	LG faults
Mismatch percentages (%)	16.66, 33.33, 50	50, 66.67, 83.33
Fault impedance values (Ω)	0, 5, 10, 15, 20	

Table 15

Experimental results of the proposed multilayer method.

Layer Type	Training accuracy	Confusion matrix	Testing accuracy
1	95.32 %	$\begin{bmatrix} 90 & 0 \\ 9 & 141 \end{bmatrix}$	96.25 %
2	99.67 %	$\begin{bmatrix} 30 & 0 \\ 0 & 60 \end{bmatrix}$	100 %
3	100 %	$\begin{bmatrix} 30 & 0 \\ 0 & 30 \end{bmatrix}$	100 %
4	96.17 %	$\begin{bmatrix} 57 & 3 & 0 \\ 0 & 60 & 0 \\ 1 & 0 & 29 \end{bmatrix}$	97.33 %
5	99.06 %	$\begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{bmatrix}$	100 %
6	100 %	$\begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{bmatrix}$	100 %

(Amiri, Kichou, et al., 2024), and (Yahyaoui et al., 2023), which have attempted to provide an accurate model to detect and classify several faults in PV arrays. However, the main drawback is that none of the above-mentioned studies have taken into consideration the critical fault detection conditions. Therefore, the process of fault detection has become very easier to carry out. Also, their models are not capable of any severity assessment to provide an accurate output for protection devices. Hence, although the models might apparently yield relatively high accuracies, they do not seem to be reliable in real-world conditions.

b) In several studies, such as (Dhimish & Tyrrell, 2023) and (Amiri, Oudira, et al., 2024), authors have positively presented models which have considered a difficult condition to detect the faults (usually critical mismatch levels). In addition, the models are capable to assess the severity of some faults in PV arrays. But there seems to be other problems, such as in (Dhimish & Tyrrell, 2023), the model is incomprehensive and covers only one single type of faults and two classes in the dataset. It also seems that due to the consideration of a challenging fault detection condition, the model is not able to produce sufficient accuracy therefore it is argued to be unreliable in practice. In addition, in (Amiri, Oudira, et al., 2024), in order to provide a high accuracy, the model has to be fully trained using a massive dataset which causes the above-mentioned problems.

c) A large number of fault scenarios is also provided in (Hojabri et al., 2022), (Hajji et al., 2023), and (Yahyaoui et al., 2023), along with plenty of classes in the dataset. However, various disadvantages can be witnessed such as the model presented in (Hojabri et al., 2022) does not yield a sufficient accuracy, has not considered the critical fault detection conditions, and is not able to assess the severity of faults in PV arrays. Moreover, although the proposed models in (Hajji et al., 2023), and (Yahyaoui et al., 2023) are producing a seemingly satisfying accuracy, they require a huge number of data samples, has not taken into consideration the critical fault detection conditions, and are not capable of severity assessment.

d) The model presented in (Kumari & Panigrahi, 2023) seems to be address various challenges. For instance, fault impedance, as another important factor (added to critical mismatch levels) which adversely affects the severity of faults thus imposing a huge burden on fault

Table 16

Results of a comparison between the proposed model and various recent models in literature.

References	Fault scenarios	Classes	Utilized technique	Critical fault detection conditions	Severity Assessment	Training data samples	Avg. Experimental Accuracy
(Hojabri et al., 2022)	- Connector faults - PID faults - Partial shading - Building shadow - SCBD - Partial soiling - Glass breakage	9	ANN	Neither	No	4110	93 %
(Hajji et al., 2023)	Line-to-line faultsLine-to-ground faultsBypass diode faultsConnectivity faults	11	Various neural networks	Neither	No	400,000	100 %
(Dhimish & Tyrrell, 2023)	Bypass diode faults	2	ANN	Mismatch only	Yes	1680	93.6 %
(Yahyaoui et al., 2023)	- Line-to-line faults - Line-to-ground faults - Bypass diode faults - Connectivity faults - Line-to-line faults - Line-to-ground faults	11	One-class classification (various single classifiers)	Neither	No	350,000	99.68 %
(Kumari & Panigrahi, 2023)	- Line-to-line faults - Line-to-ground faults	6	GA + Weighted kNN	Impedance Mismatch	Yes	1478	95.18 %
(Amiri, Kichou, et al., 2024)	- Line-to-line faults - Inverter disconnection - Partial shading	6	CNN + Bi-GRU	Neither	No	437,202	100 %
(Amiri, Oudira, et al., 2024b)	- Line-to-line faults - Open-circuit faults - Partial shading	5	Random forest	Mismatch only	Yes	30,214	99.4 %
Proposed method in this study	- Line-to-line faults - Line-to-ground faults - Open-circuit faults - Degradation faults	10	Weighted ensemble learning model	Impedance Mismatch	Yes	1392	98.37 %

PID: Potentially induced degradation, SCBD: Short circuit in blocking diode, ANN: Artificial neural network, GA: Genetic algorithm, CNN: Convolutional neural network, Bi-GRU: Bidirectional gated recurrent unit.

detection process, is considered in (Kumari & Panigrahi, 2023). They have also taken into account the ability of severity assessment, and a small number of data samples to train their model. However, the main drawback in (Kumari & Panigrahi, 2023) is that the rather medium-to-low accuracy of the model, especially when dealing with less severe faults cannot be neglected at all.

e) Finally, looking closely at the proposed multilayer model in this study, it is clear that it attempts to overcome all mentioned challenges in previous studies and outperforms the other models from various aspects at the same time; (i) the proposed model in this study covers a large number of electrical faults in PV arrays, (ii) the proposed model has considered the most difficult fault detection conditions (both critical impedance values and critical mismatch levels), (iii) the proposed multilayer model is capable of assessing the severity of faults, (iv) the model requires only a small number of data samples to train the model and even shrinks the dataset more as the dataset is split into several subsets in each layer, and (v) the proposed model is thus able to provide a high reliable accuracy.

6. Future work and conclusions

6.1. Limitations and future work

The area of fault diagnosis in PV arrays can be comprehensive including three different procedures; detection, classification, and localization. The first two are dealt with in this study by fully addressing the existing challenges and problematic conditions. However, the process of fault localization especially using sensorless state-of-the-art techniques is pending for future and has not been sufficiently considered in literature. It is worth mentioning that the present studies have proposed several models to pinpoint the precise location of a fault occurrence but almost all of them utilize a large number of expensive sensors and actuators which do not seem economically efficient in real-

world conditions.

6.2. Conclusions

Fault detection plays a pivotal role in improving the performance and service life of PV systems. In recent years, several techniques have been proposed to detect and classify electrical faults in PV systems. In this study, an intelligent and autonomous fault detection and classification method is presented. First, several faulty and normal data samples were created through a real PV array. Then, a multilayer model was presented which consisted of six layers: (1) distinguishing between zero-to low-current and high-current faults, (2) precise detection of normal conditions compared to OC and string degradation faults, (3) differentiation between OC and string degradation faults, (4) classifying LL, LG, and array degradation faults, and (5) & (6) LL and LG faults severity detection, respectively, with respect to several mismatch levels. During the process of feature extraction, sixteen features were elicited from the PV array I-V curve under faulty and normal conditions. In addition, to reduce the dimensionality of the dataset, thus simplifying the training process, the SFFS algorithm was applied as a feature selection technique in each layer to select the most effective features for the detection and classification process. Finally, a WEL algorithm which included three single algorithms namely SVM, NB and LR was utilized in each layer to detect and classify the electrical faults. Note that the weight optimization process was carefully fulfilled by a GA to obtain the optimal weights for each single classifier in the WEL algorithm. The model features high accuracy, comprehensive fault detection and classification, and severity identification of various electrical faults in PV arrays. It was tested on a stand-alone PV array and the results demonstrated promising outcomes for detection, classification and severity identification of the electrical faults with the average simulation and experimental accuracy of 98.9 % and 98.37 %, respectively.

CRediT authorship contribution statement

Aref Eskandari: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Visualization, Writing – review & editing. **Amir Nedaei:** Formal analysis, Methodology, Visualization, Writing – original draft. **Jafar Milimonfared:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Supervision, Validation. **Mohammadreza Aghaei:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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